IMPORT DATA USING TRANSFORM MAPS (Spreadsheet)

A Project Report Submitted To

BHARATHIDASAN UNIVERSITY, TRICHIRAPPALLI

In Partial Fulfillment of the Requirements for the Award of Degree of

BACHELOR OF COMPUTER APPLICATIONS

Submitted by

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IMAYAM ARTS AND SCIENCE COLLEGE,

KANNANUR – 621206, THURAIYUR.

SEPTEMBER – 2026

Import Data using Transform Maps (Spreadsheet)

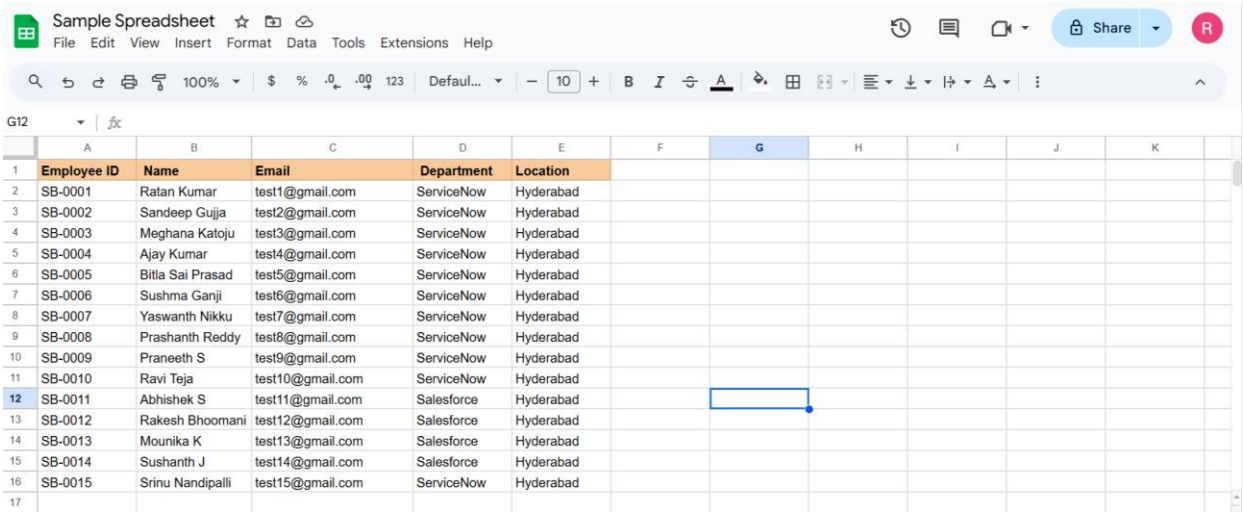
Description:

This micro project focuses on importing structured data from an external spreadsheet into the **ServiceNow** platform using Import Sets and Transform Maps. The project simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields. Upon executing the transformation, records are created in the target table and validated to ensure data integrity. This project helps beginners understand the end-to-end data import process in ServiceNow, including data loading, field mapping, transformation execution, and result verification.

Step by Step process:

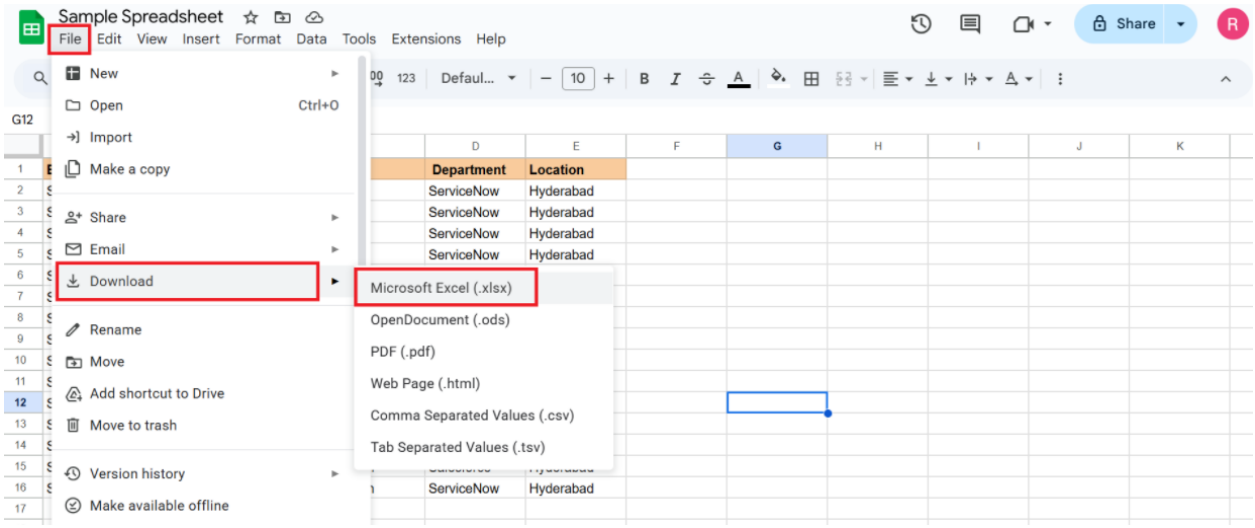
Phase 1: Prepare the sheets

1. Open New Google Spreadsheet
2. Create New Google Spreadsheet with sample data as shown below



	A	B	C	D	E	F	G	H	I	J	K
1	Employee ID	Name	Email	Department	Location						
2	SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad						
3	SB-0002	Sandeep Gujja	test2@gmail.com	ServiceNow	Hyderabad						
4	SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad						
5	SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad						
6	SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad						
7	SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad						
8	SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad						
9	SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad						
10	SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad						
11	SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad						
12	SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad						
13	SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad						
14	SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad						
15	SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad						
16	SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad						
17											

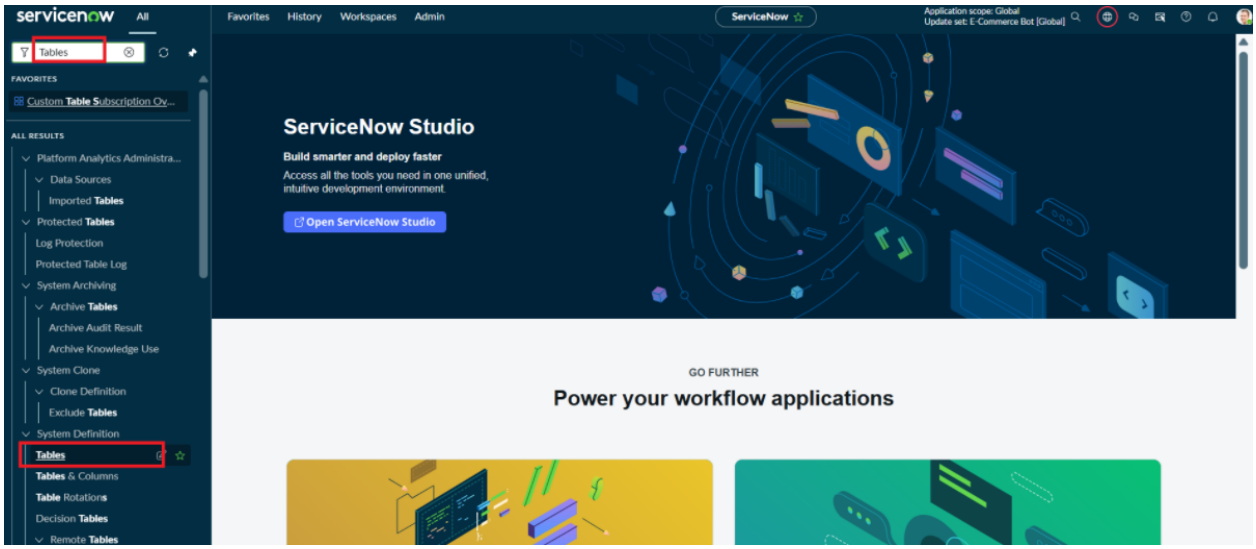
3. Download the spreadsheet as Sample Spreadsheet.xlsx format.



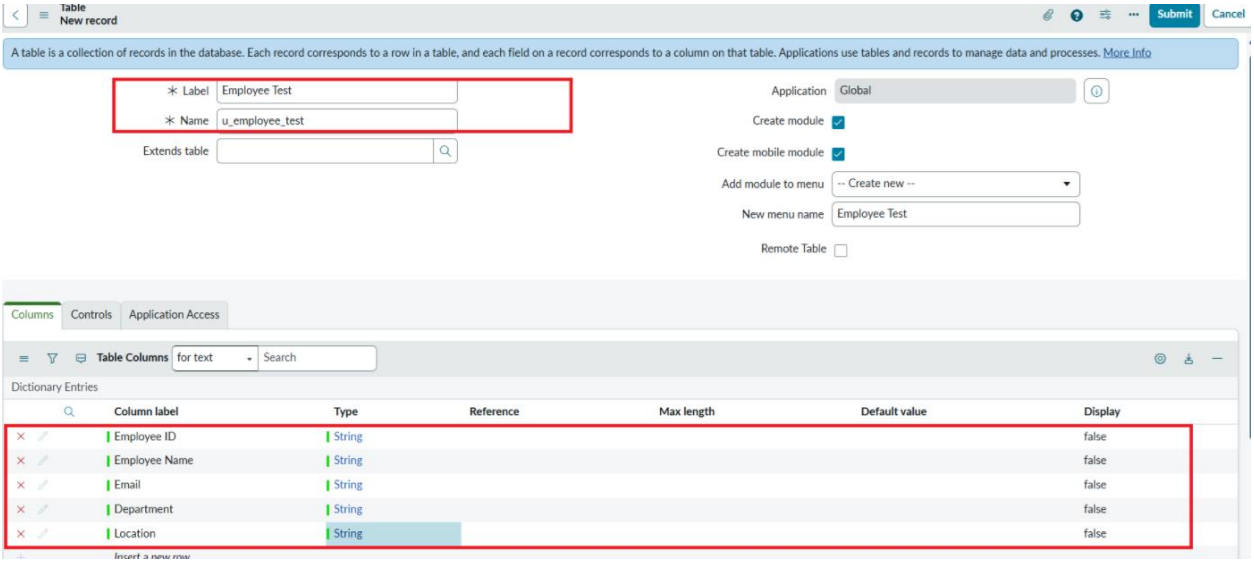
Phase 2: Create new Custom table

Navigation:

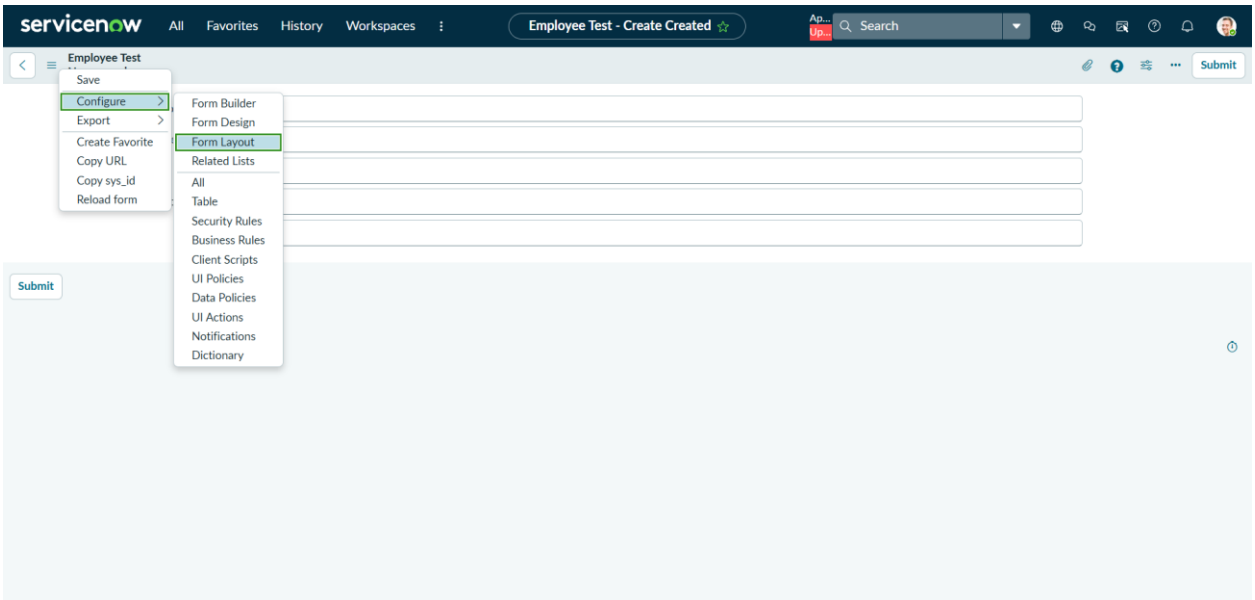
- Tables > Create New



- Label: Employee Test
- Name: u_employee_test
- After creating and saving the **Employee Test** table, **scroll down** to the bottom of the table form.



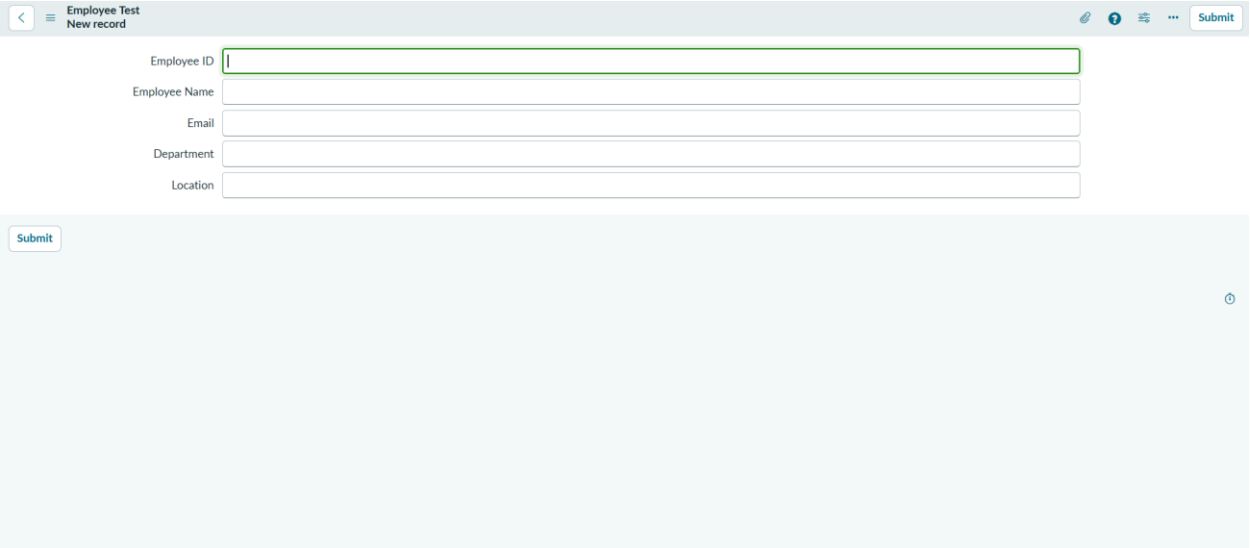
- Click **Show Form** to open the table form.
- On the form, locate the **Form Context Menu / Additional Actions** option on the left side of the form header.
- Click the **Form Context Menu / Additional Actions** icon.
- Click configure , select **Form Layout**.



- The **Form Layout** page will open, where you can configure the fields displayed on the form
- After opening **Form Layout**, click **New** to create a new field.
- Enter the field details:
- **Column Label:** Employee ID
- **Type:** String
- Click **Add** to create the field.

- Repeat the same steps to create the following fields:
    - **Employee Name** – Type: String

    - **Email** – Type: String
    - **Department** – Type: String
    - **Location** – Type: String
- After creating all the required fields, click **Save**.
- Verify that the newly created fields are displayed on the **Employee Test** form.

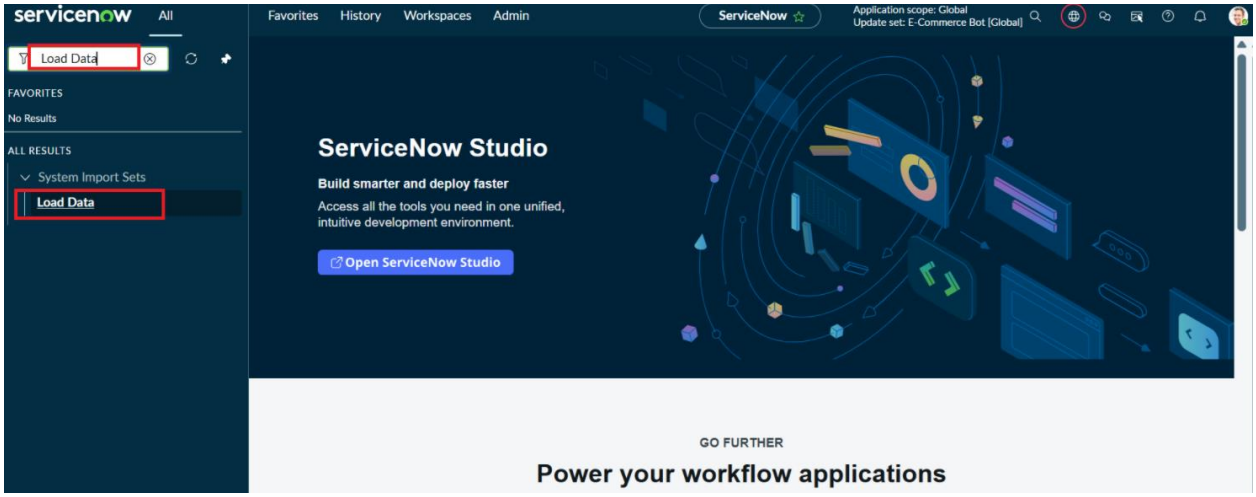


Click on Submit or Save.

Phase 3: Import Set Table

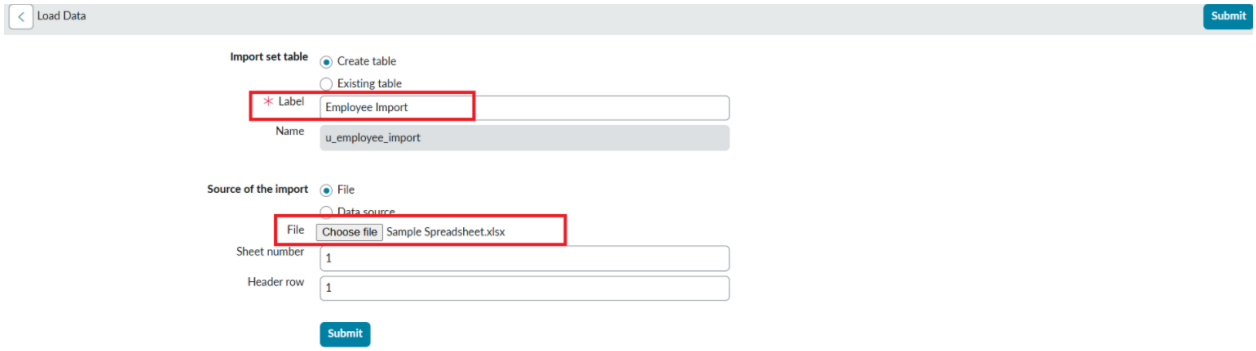
Navigation:

In Application Navigator → Load Data



Click on Load Data and create an import table
Fill details as shown below

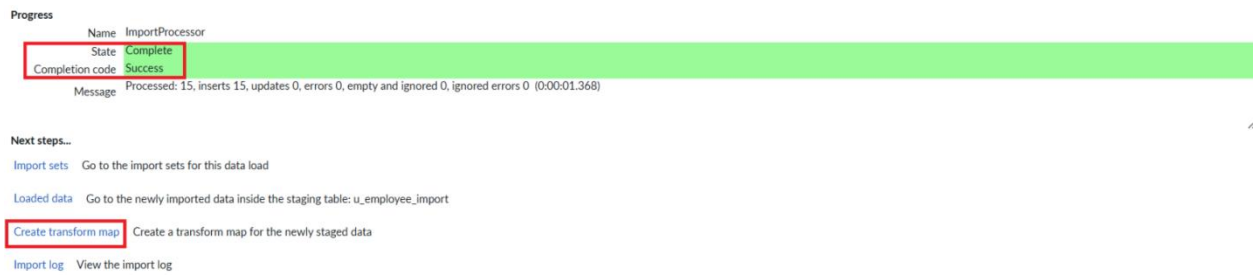
- Label: Employee Import
- Name: u_employee_import (auto populated)



- Click on Submit

Once we click on Submit, we can see the success message as shown below.

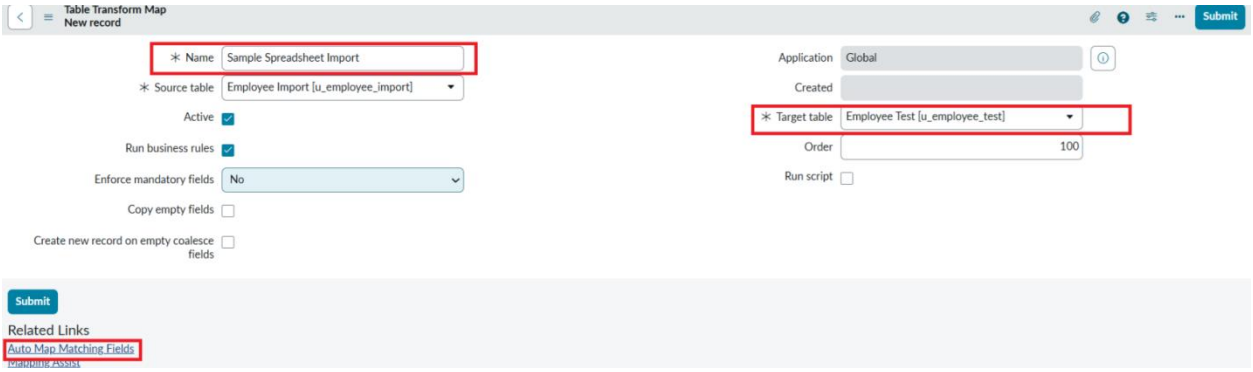
Also we need to make sure to click on Create Transform Map.



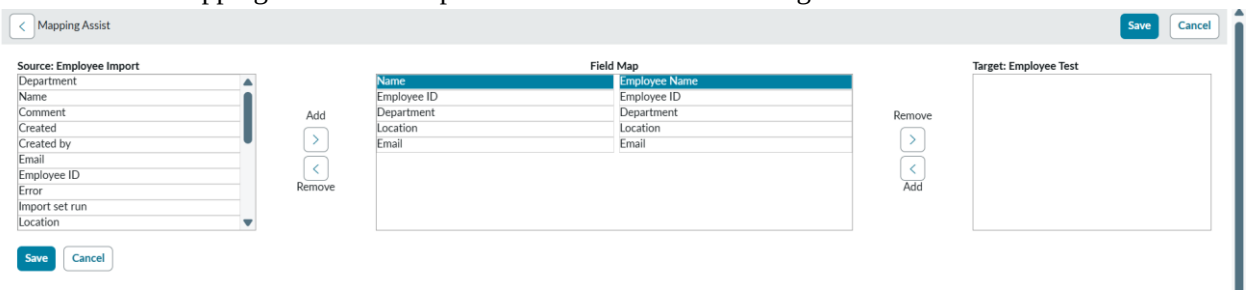
Phase 4: Create Transform Map

In this phase we need to create the Transform Map by clicking on the Create Transform Map button as shown in above screenshot.

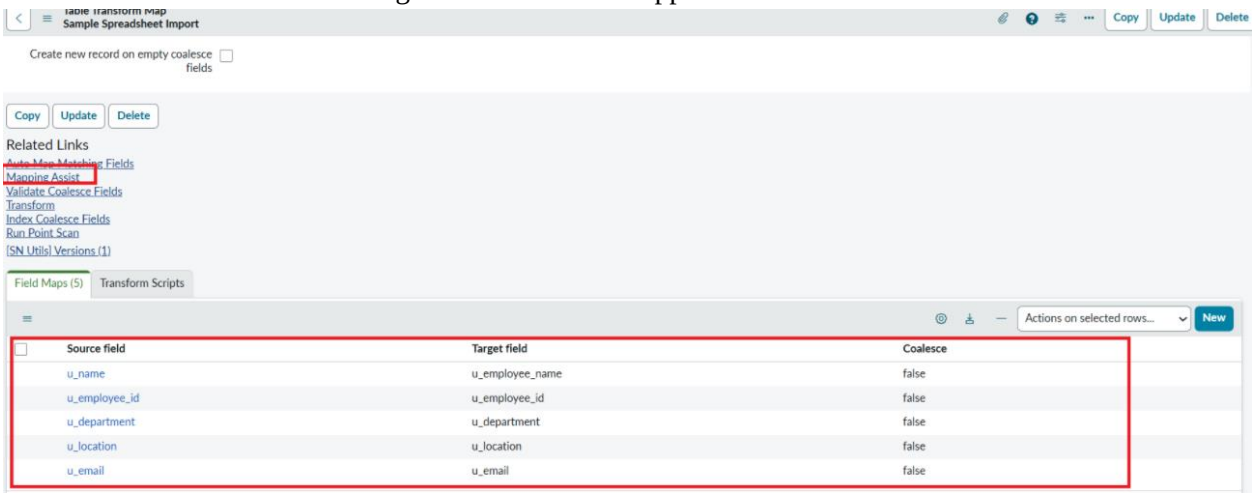
Give inputs as shown below.
Name: Sample Spreadsheet Import
Source Table: Employee Import is auto-populated
Target Table: Employee Test - as we created earlier
Click on Auto Map Matching fields.



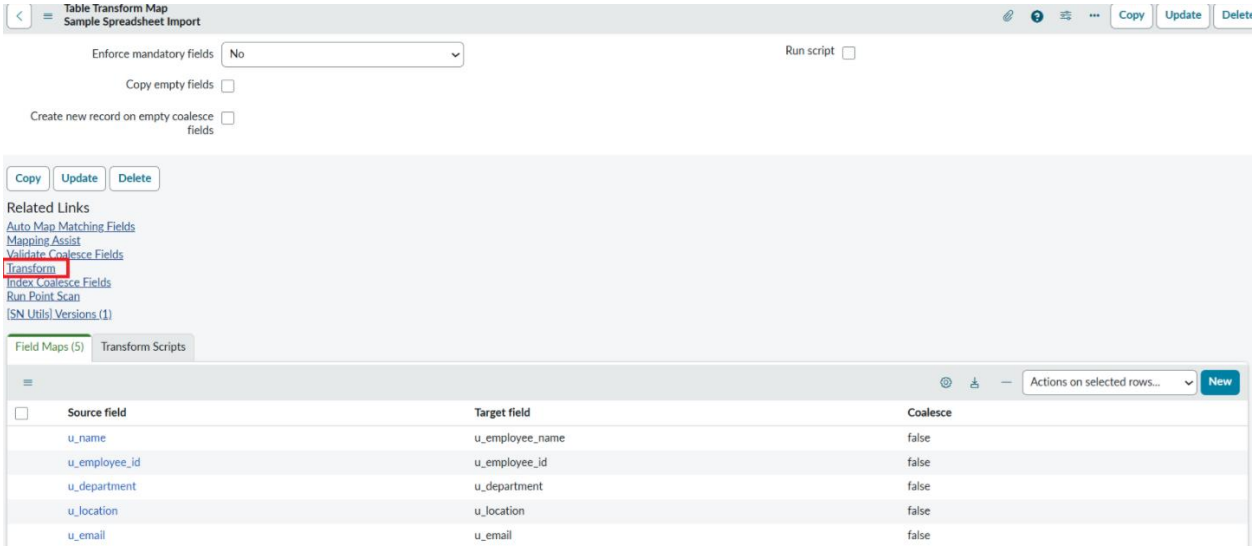
Now click on Mapping Assist and map the fields with Source & Target Table



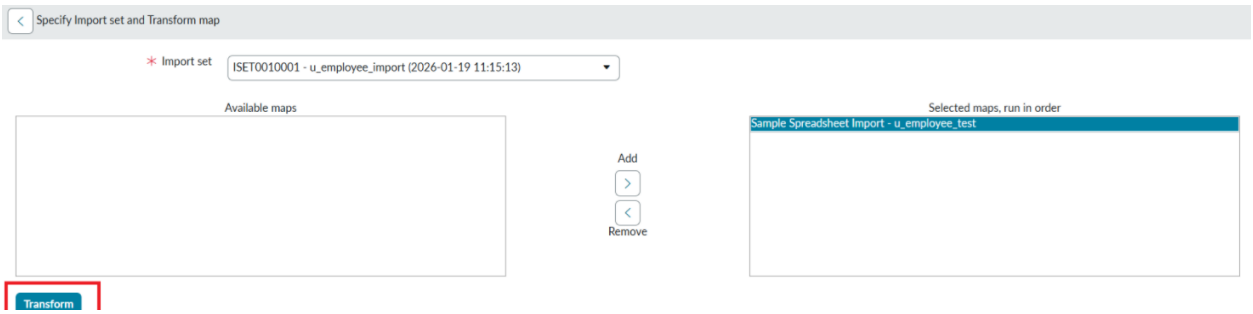
Click on Save
Make sure the both Source & Target table fields are mapped as shown below.



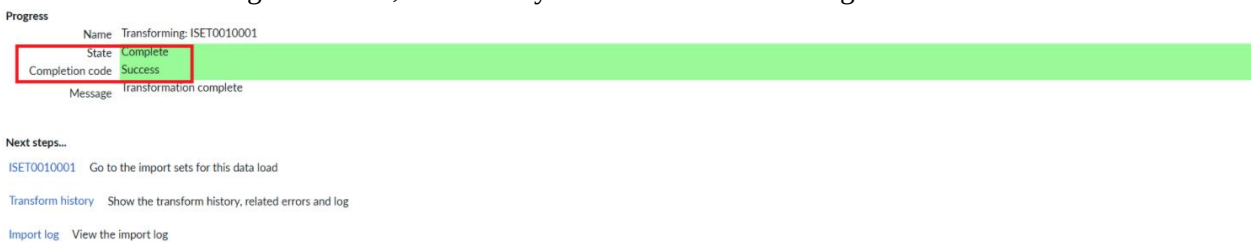
- Now click on Transform as shown below



- Now click on Transform Button as shown below.



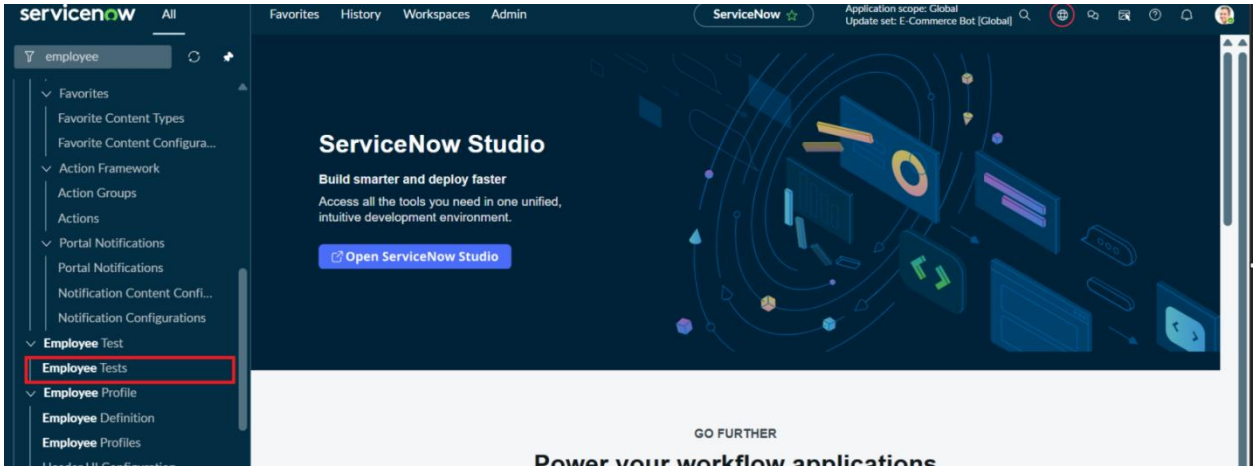
- After clicking Transform, make sure you see the success message as shown below.



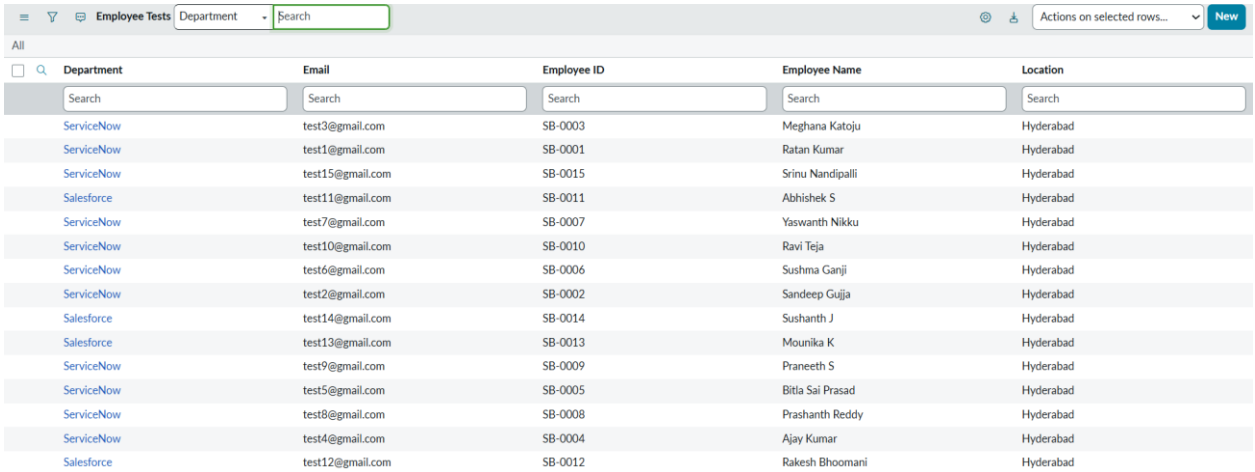
Phase 5: Transform Data & Validate

Navigation:

- Open Employee Test Table from application navigator as shown below.

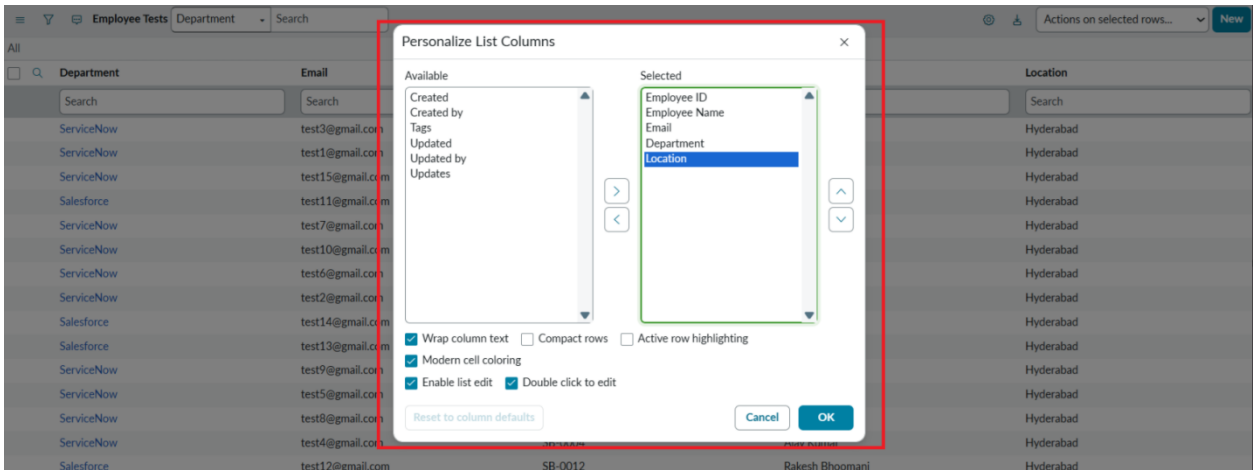


- As our import got succeeded, all the records are created in the Employee Test Table.



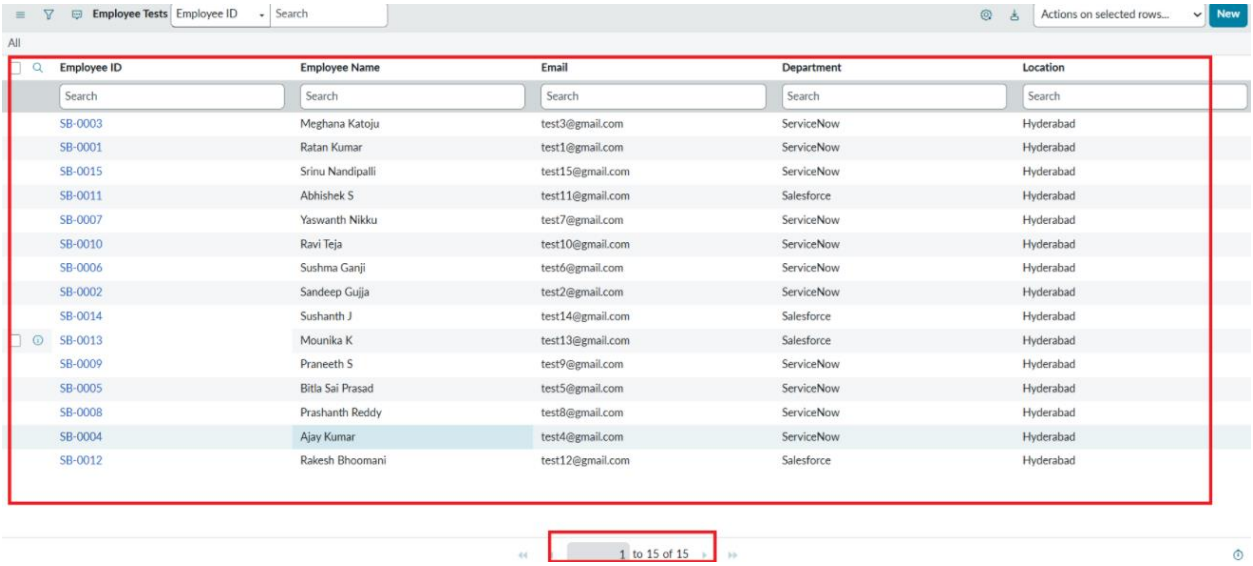
Department	Email	Employee ID	Employee Name	Location
ServiceNow	test3@gmail.com	SB-0003	Meghana Katoju	Hyderabad
ServiceNow	test1@gmail.com	SB-0001	Ratan Kumar	Hyderabad
ServiceNow	test15@gmail.com	SB-0015	Srinu Nandipalli	Hyderabad
Salesforce	test11@gmail.com	SB-0011	Abhishek S	Hyderabad
ServiceNow	test7@gmail.com	SB-0007	Yaswanth Nikku	Hyderabad
ServiceNow	test10@gmail.com	SB-0010	Ravi Teja	Hyderabad
ServiceNow	test6@gmail.com	SB-0006	Sushma Ganji	Hyderabad
ServiceNow	test2@gmail.com	SB-0002	Sandeep Gujja	Hyderabad
Salesforce	test14@gmail.com	SB-0014	Sushanth J	Hyderabad
Salesforce	test13@gmail.com	SB-0013	Mounika K	Hyderabad
ServiceNow	test9@gmail.com	SB-0009	Praneeth S	Hyderabad
ServiceNow	test5@gmail.com	SB-0005	Bitla Sai Prasad	Hyderabad
ServiceNow	test8@gmail.com	SB-0008	Prashanth Reddy	Hyderabad
ServiceNow	test4@gmail.com	SB-0004	Ajay Kumar	Hyderabad
Salesforce	test12@gmail.com	SB-0012	Rakesh Bhoomani	Hyderabad

- Make sure you rearrange the fields accordingly by using Personalised List Columns



Click on OK

Final Result:
All the records are successfully imported into the ServiceNow system with Import Set, Transform Map.

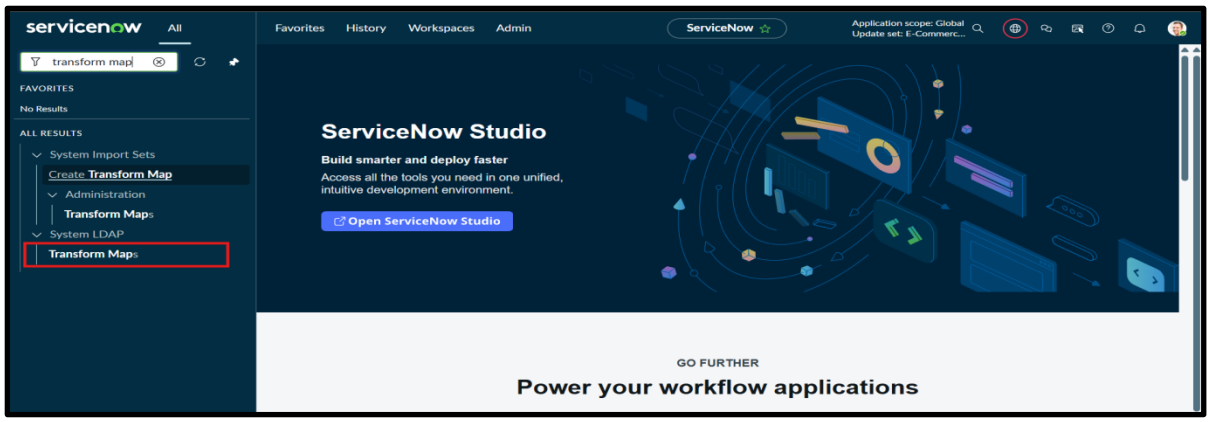


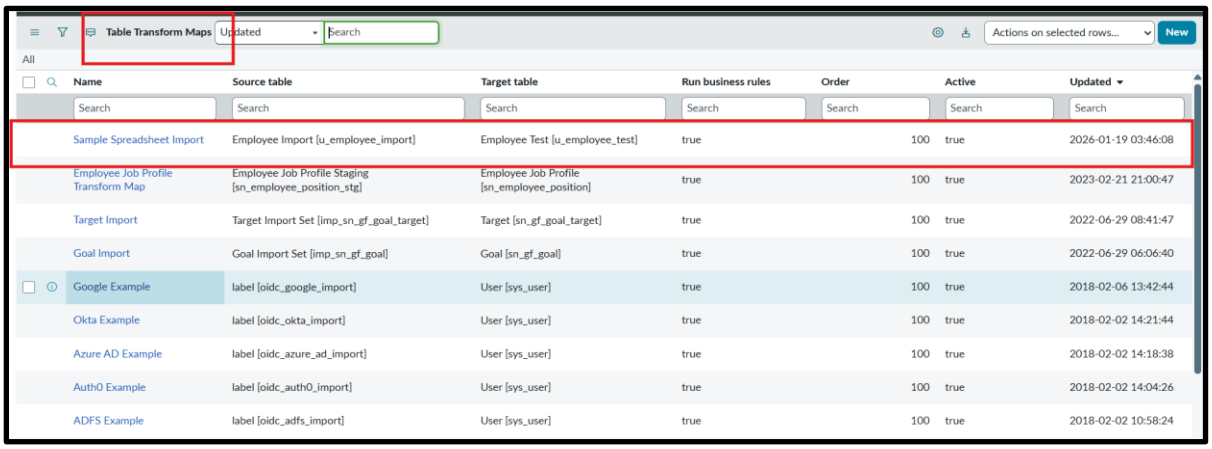
Phase 6: Enable Coalesce to Avoid Duplicate Records

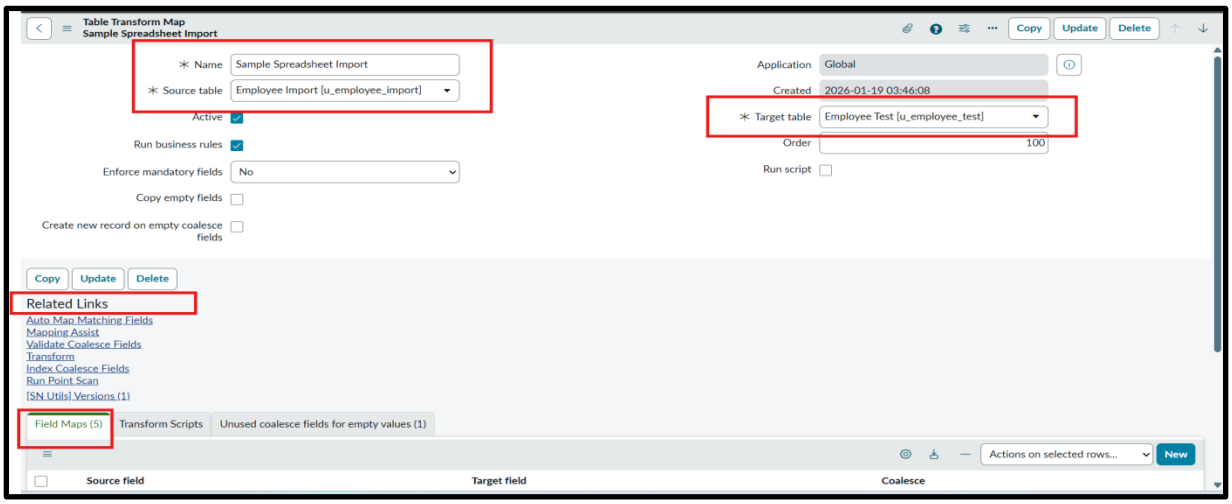
- In real projects, the same employee data may be imported multiple times.
- To prevent duplicate employee records, we use **Coalesce**.
- The Coalesce will update the same records with new values if we add and the duplicate values are not inserted.

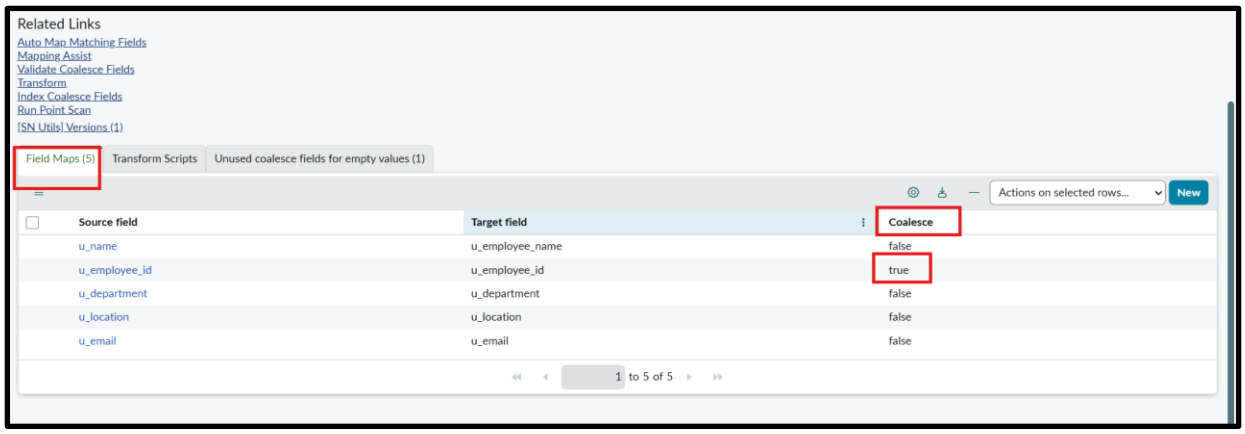
Steps:

1. Open Transform map Under System LDAP On All menus.
2. Open the Created Transform map , Name is Sample Spreadsheet Import.
3. Under Related list, select the Field maps Section and Make any one field Coalesce True which appears in False.
4. Now save the form.
5. Then again add some data which is in Excel format.


servicenow
All
transform map
FAVORITES
No Results
ALL RESULTS
System Import Sets
Create Transform Map
Administration
Transform Maps
System LDAP
Transform Maps
Favorites
History
Workspaces
Admin
ServiceNow
Application scope: Global
Update set: E-Commerc...
ServiceNow Studio
Build smarter and deploy faster
Access all the tools you need in one unified, intuitive development environment.
Open ServiceNow Studio
GO FURTHER
Power your workflow applications


Table Transform Maps
Updated
Search
Actions on selected rows...
New
All
Name
Source table
Target table
Run business rules
Order
Active
Updated
Sample Spreadsheet Import
Employee Import [u_employee_import]
Employee Test [u_employee_test]
true
100
true
2026-01-19 03:46:08
Employee Job Profile Transform Map
Employee Job Profile Staging [sn_employee_position_stg]
Employee Job Profile [sn_employee_position]
true
100
true
2023-02-21 21:00:47
Target Import
Target Import Set [imp_sn_gf_goal_target]
Target [sn_gf_goal_target]
true
100
true
2022-06-29 08:41:47
Goal Import
Goal Import Set [imp_sn_gf_goal]
Goal [sn_gf_goal]
true
100
true
2022-06-29 06:06:40
Google Example
label [oidc_google_import]
User [sys_user]
true
100
true
2018-02-06 13:42:44
Okta Example
label [oidc_okta_import]
User [sys_user]
true
100
true
2018-02-02 14:21:44
Azure AD Example
label [oidc_azure_ad_import]
User [sys_user]
true
100
true
2018-02-02 14:18:38
Auth0 Example
label [oidc_auth0_import]
User [sys_user]
true
100
true
2018-02-02 14:04:26
ADFS Example
label [oidc_adfs_import]
User [sys_user]
true
100
true
2018-02-02 10:58:24

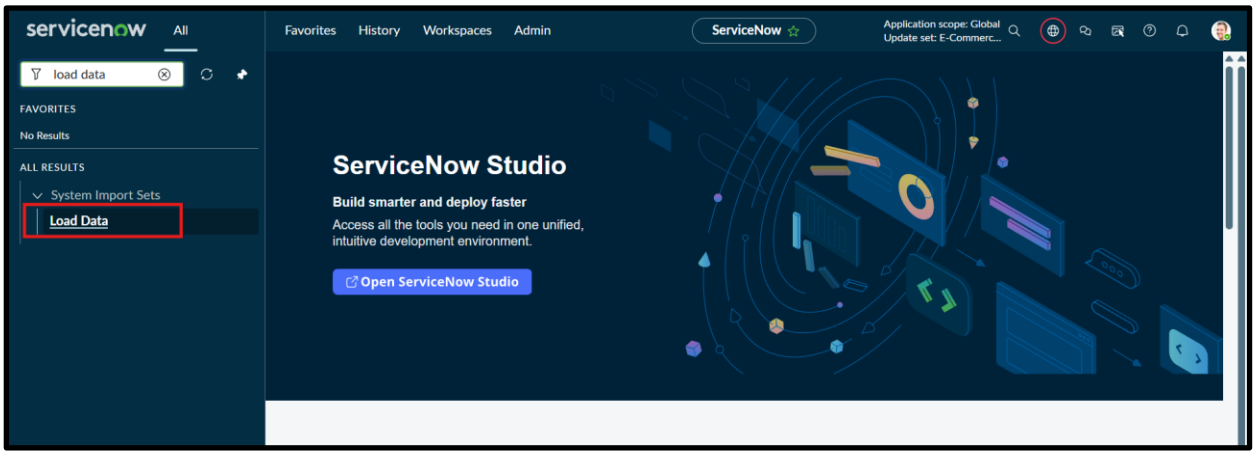

Table Transform Map
Sample Spreadsheet Import
Copy
Update
Delete
Name
Sample Spreadsheet Import
Source table
Employee Import [u_employee_import]
Active
Run business rules
Enforce mandatory fields
No
Copy empty fields
Create new record on empty coalesce fields
Application
Global
Created
2026-01-19 03:46:08
Target table
Employee Test [u_employee_test]
Order
100
Run script
Copy
Update
Delete
Related Links
Auto Map Matching Fields
Mapping Assist
Validate Coalesce Fields
Transform
Index Coalesce Fields
Run Point Scan
[SN Utils] Versions (1)
Field Maps (5)
Transform Scripts
Unused coalesce fields for empty values (1)
Actions on selected rows...
New
Source field
Target field
Coalesce



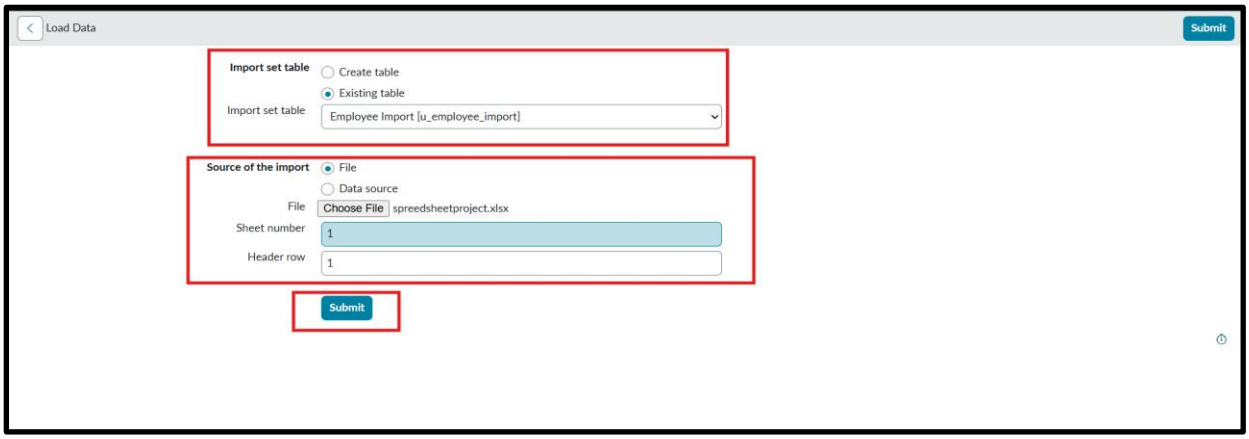
PHASE 7 : Inserting New Data In Excel Format

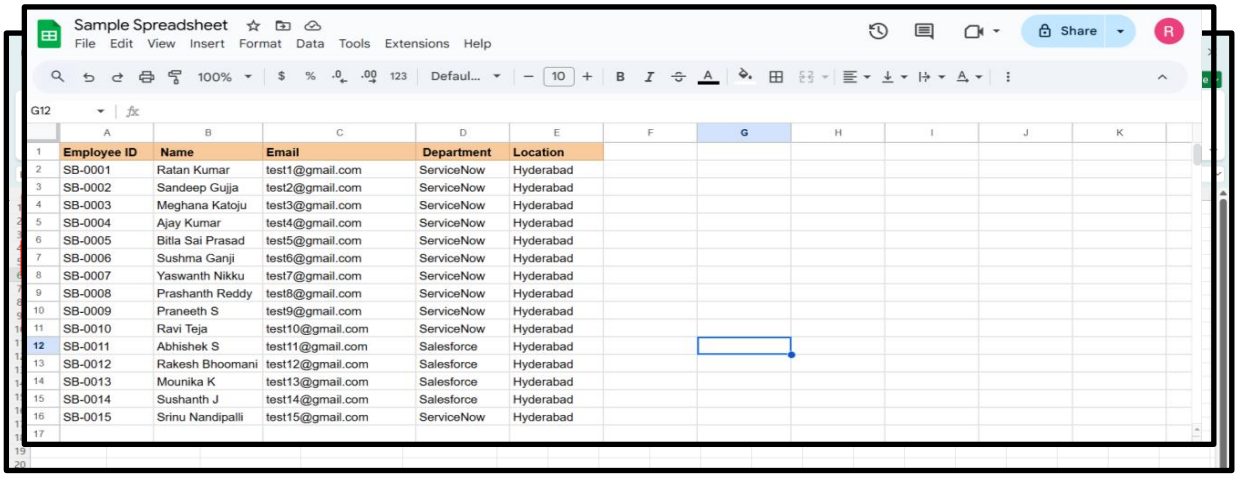
Steps :

- Navigate to All menus.
- Search for Load data Under System Import set Open Load Data.
- Fill the details as
- Import set table as select Existing table {Employee Import}.
- Source of Import is Select File and choose the file from Files.
- Sheet number : 1, Header row : 1.



- Now Click submit.



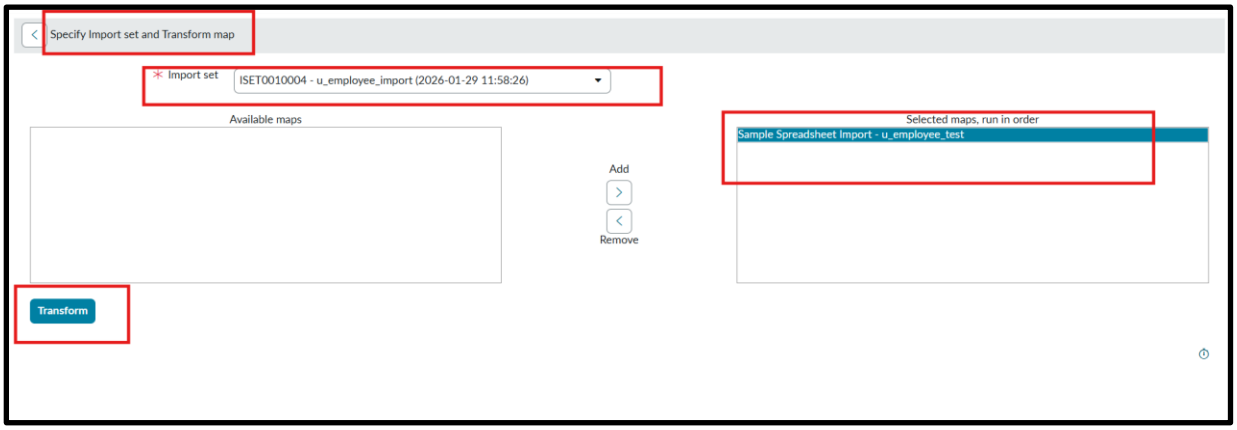


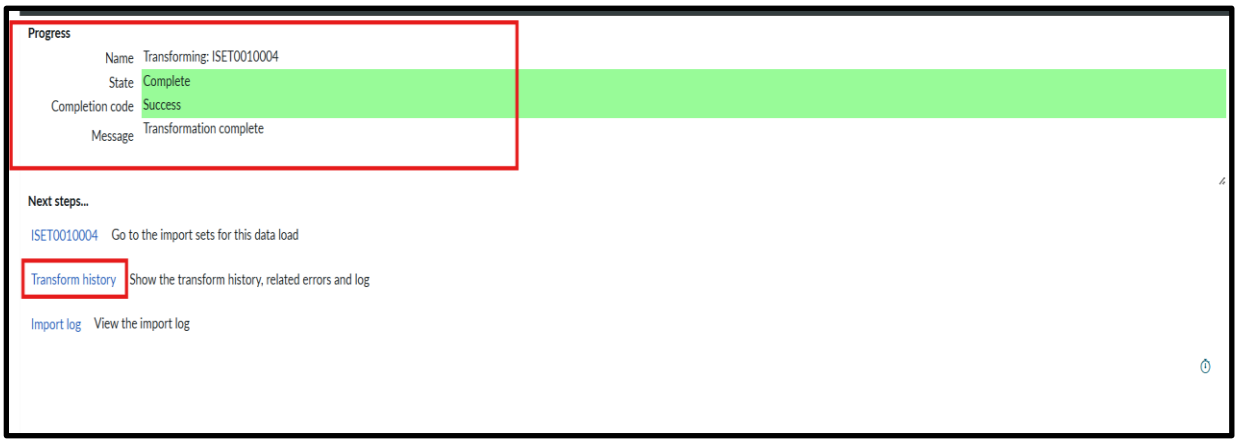
Employee ID	Name	Email	Department	Location
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Gujja	test2@gmail.com	ServiceNow	Hyderabad
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad

- Newly Created Excel Data is this.
- It is the previously added data.
- Now, In this new data we added 4 rows of the 2 rows with the same existing data with different Name changed and Email changed.
- That Raviteja Employee Email was Changed to [test10@gmail.com → test18@gmail.com].
- And Employee ID with SB-OOO4 name was changed to [Ajay kumar → Ajay].
- And two other new Employee ID’s are added.
- Now after submitting the form it appears as State Completed and Success.

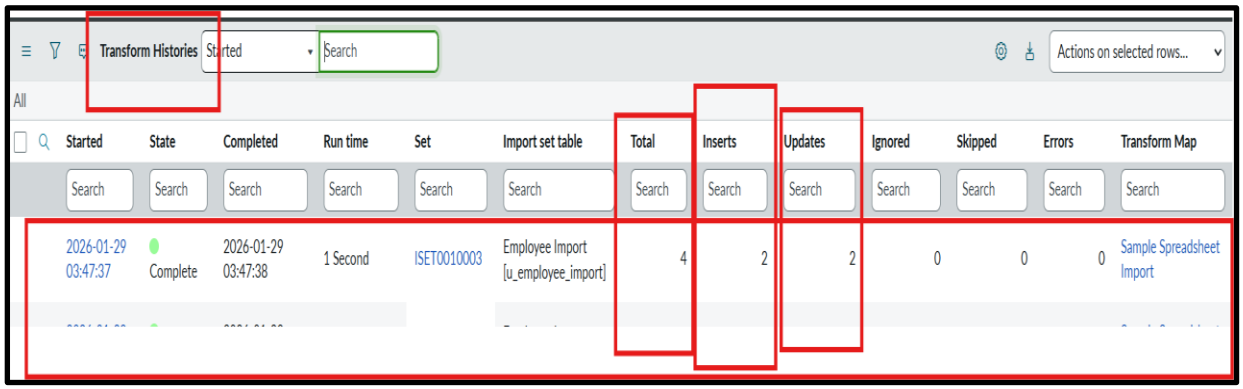


- Now Click the Run transform Under the next steps.
- Then it opens Specify Import set and transform map which is previously selected transform map then Click Transform button.

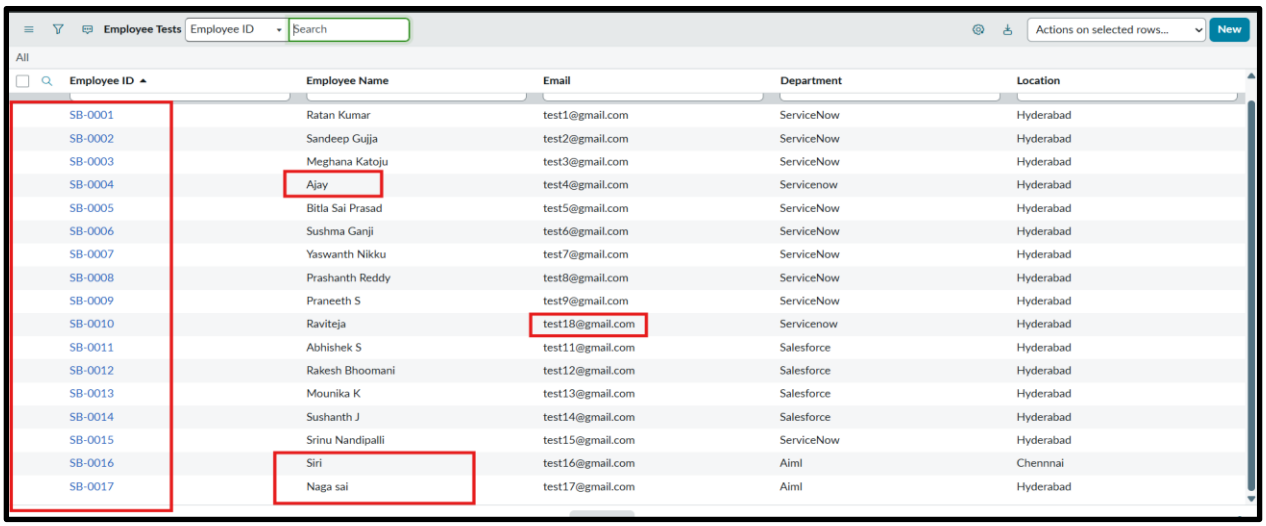




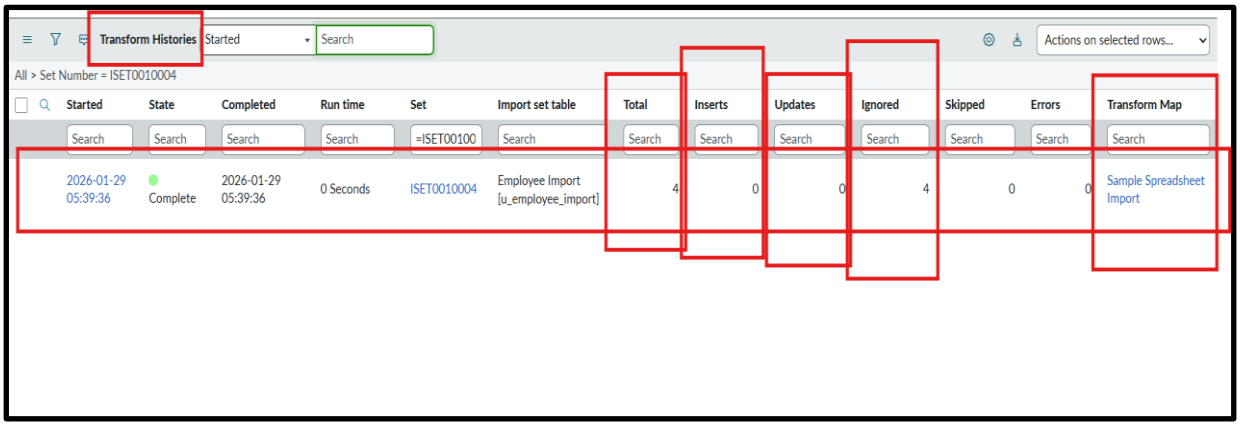
- Now the State is completed and Success.
- Now again click on the Transform history then it shows the newly added data and its status of Inserted and Skipped values out the given data.



- From the above figure : total 4 rows are uploaded out of 2 are Inserted that are two new ID's.
- And 2 rows are updates that are Same ID's with different Name and different Email that we changed.
- Then Open the Employee Tests table and see the rows updated and inserted.



- Now Repeat the Phase 7 with the same steps and same Excel sheet and Submit to check how many rows are updated and Inserted with the same data.
- Then Open the Transform history to check.



- Here we can see the total 4 are uploaded.
- On that 0 rows are Inserted and updates 0 and Ignored 4 rows.
- That same sheet we uploaded to the same transform map we created.

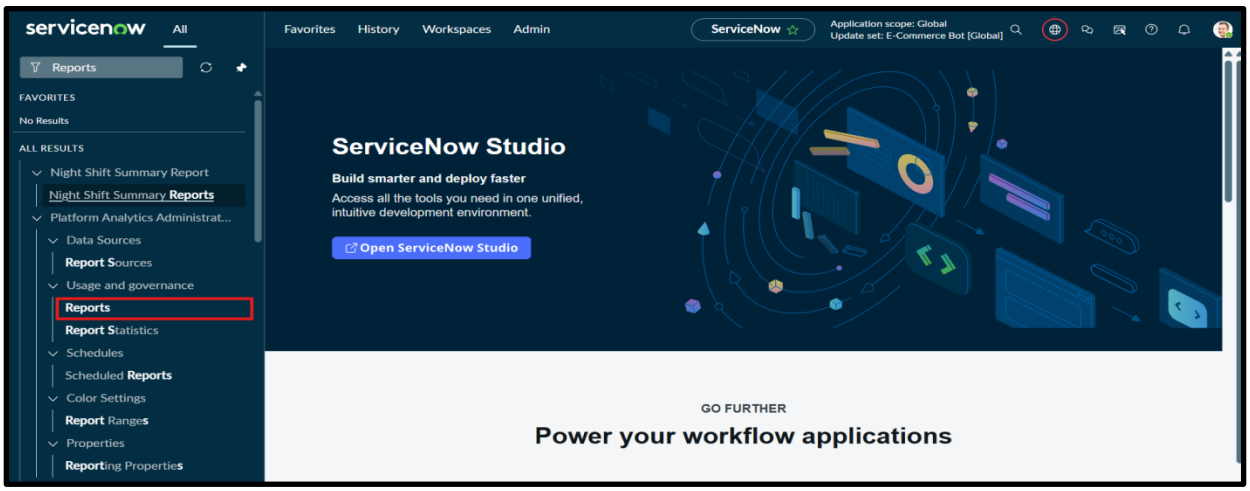
PHASE 8 : Create Reports on Employee Test Table

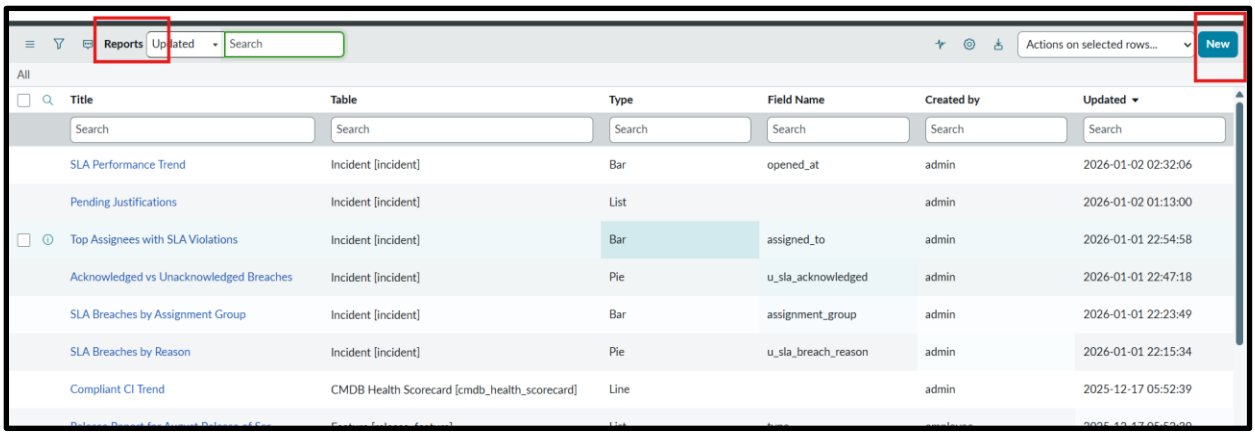
Navigation Steps :

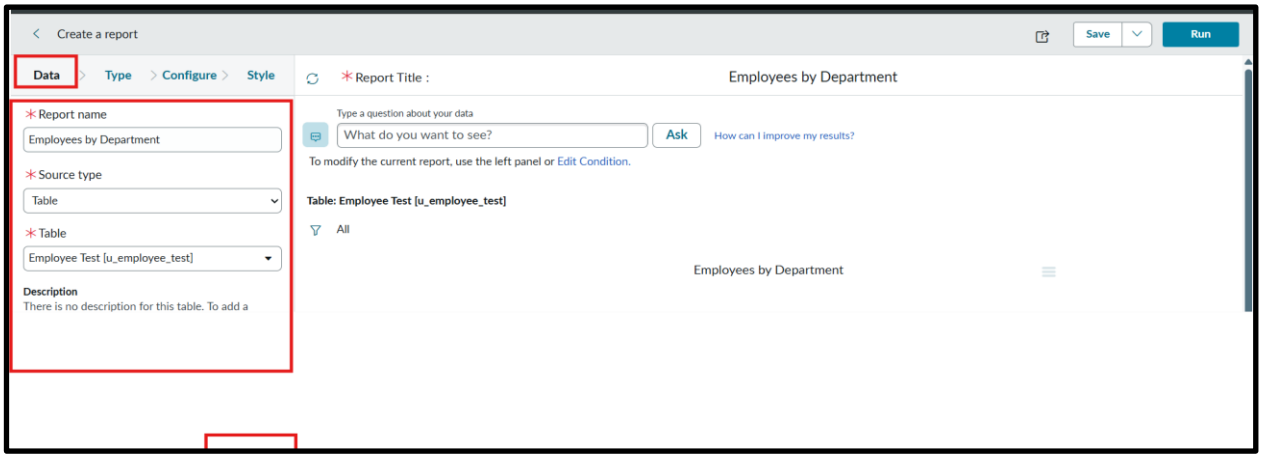
- Open All menus, Search for **Reports**.
- Under **Usage and governance** Select Reports.
- Click the **New** button to create a new report.
- Create **3 Report**s as follows.

Report 1: Employees by Department

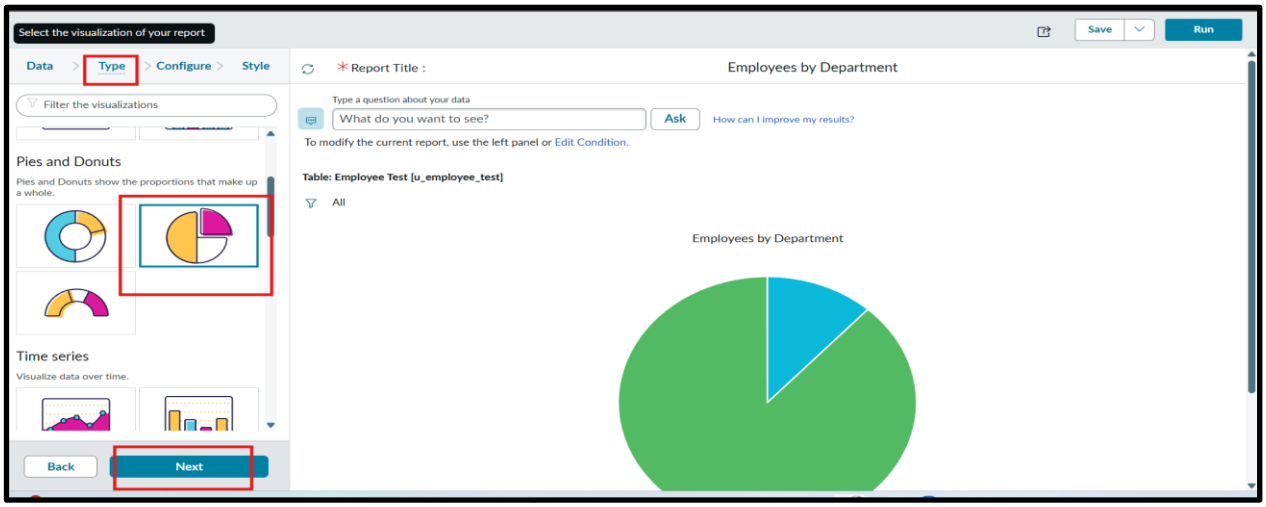
- In Data
- Report name : **Employees by Department**.
- Source Type : **Table**.
- Table : **Employee Test**, Now click the Next button.



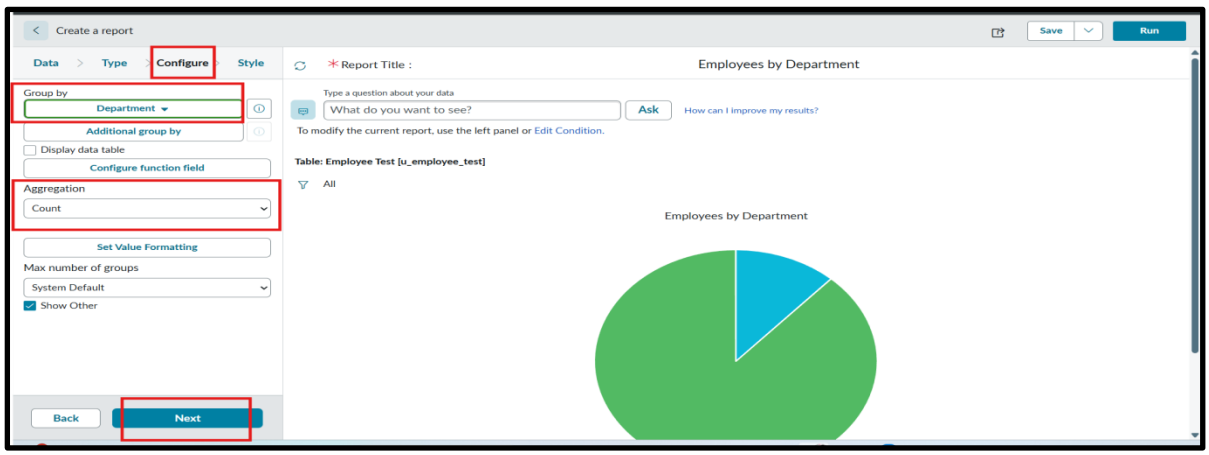




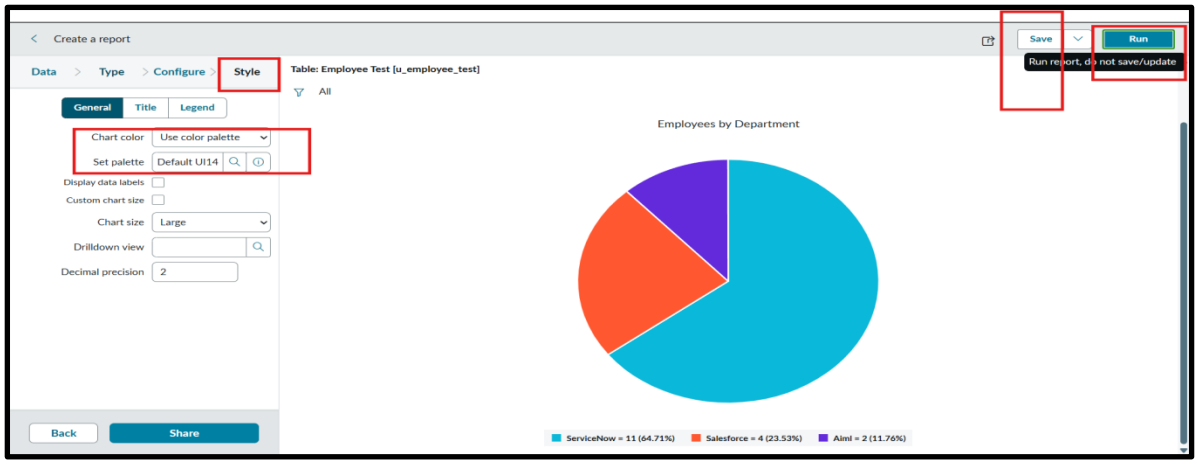
- Now after Clicking next in type Select **Pie chart** .
- And click the **next** button.



- Now In Configure Select Group by **Department** Field.
- In the Aggregation Field Select **Count.**
- Then Click **next**.



- Now In Style Leave the Chart color by **default.**
- And Click the **Run** button to Execute the given data Report.
- And Click **save** button to Save the Report.



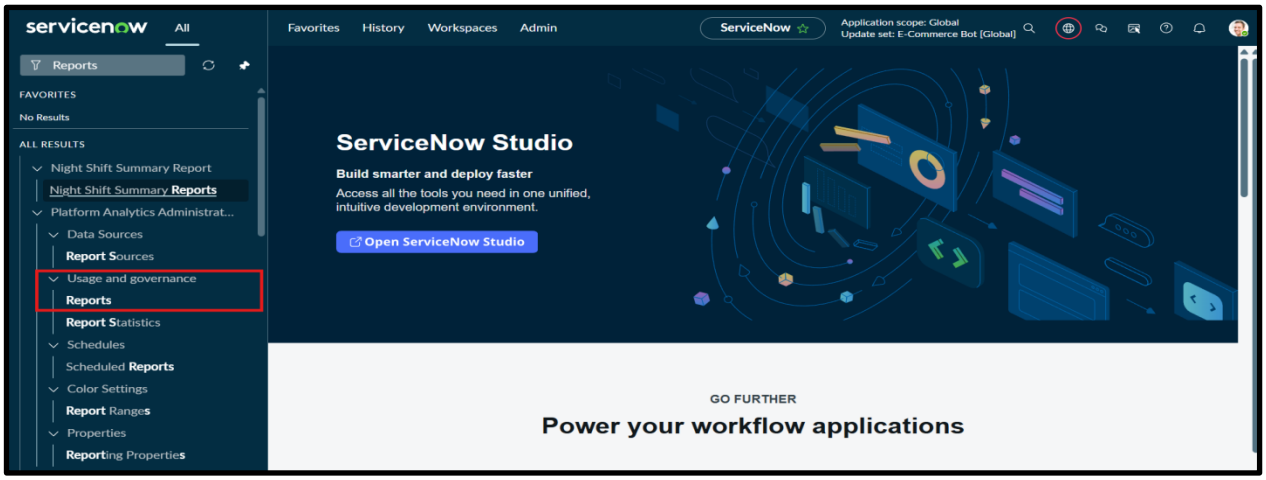
Report 2: Employees by Location

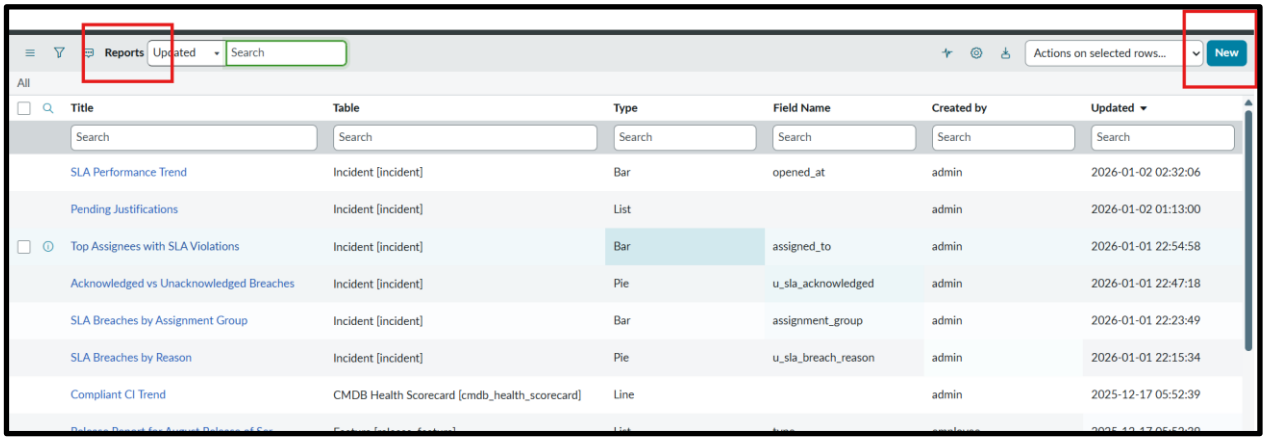
Configuration

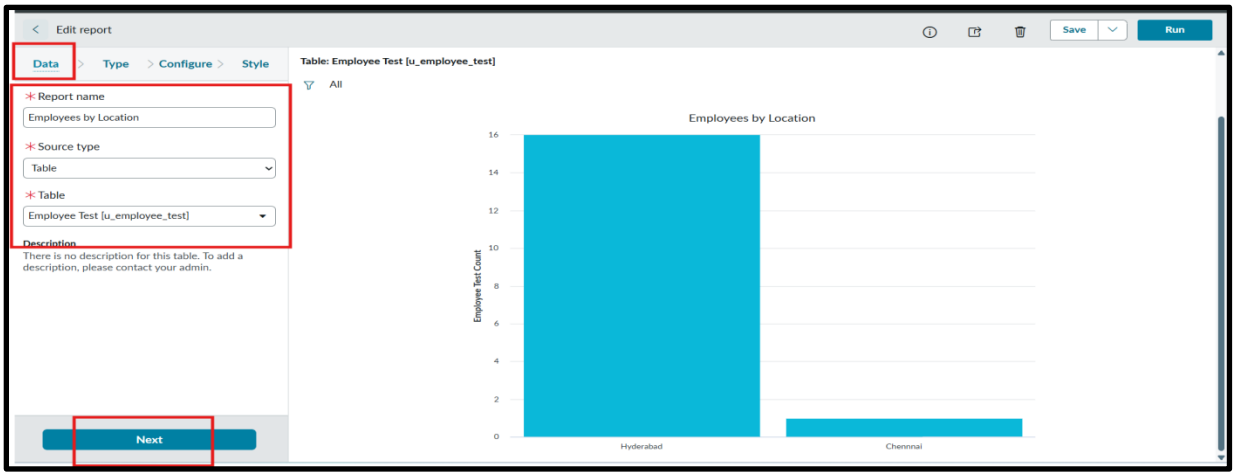
- Table: Employee Test
- Type: **Bar Chart**
- Aggregation: **Count**
- Group By: **Location**

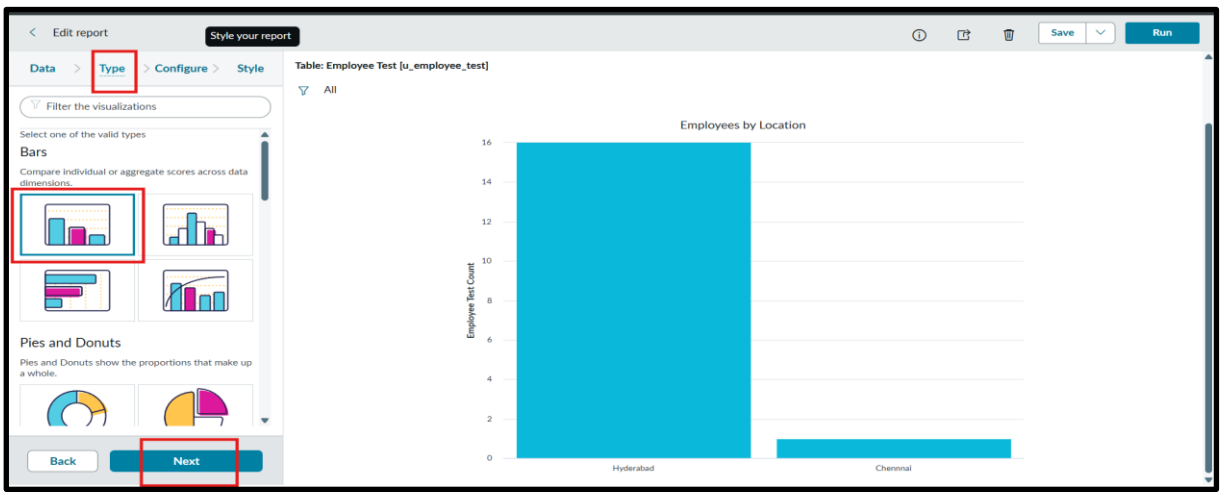
Steps :

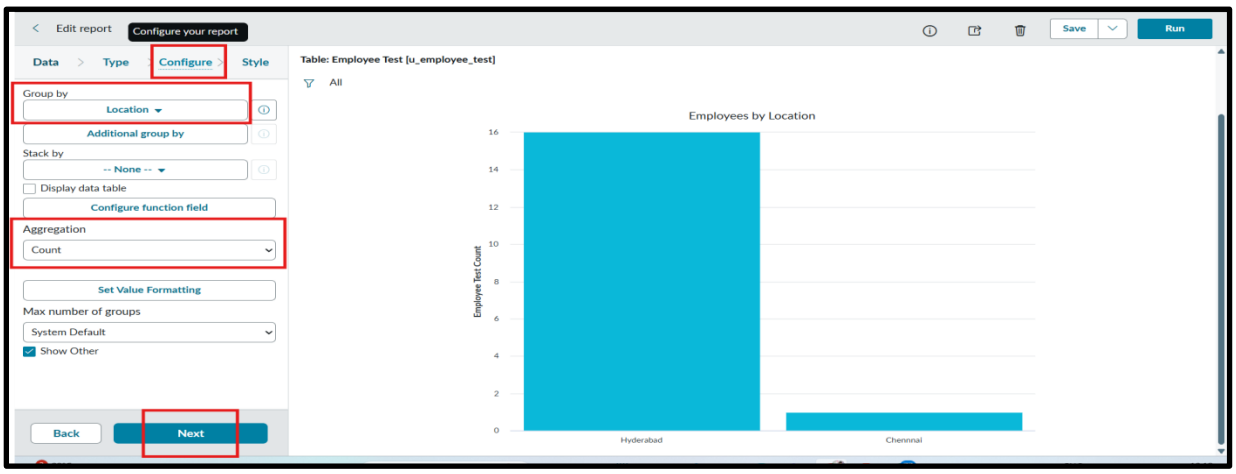
1. Create a New Report.
2. Report name : **Employees by Location.**
3. Table Select : **Employee Test**.
4. Select Type : **Bar Chart.**
5. In Configure Select Group by Field : **Location.**
6. And Aggregation is **Count**.
7. Now Click **Run** to Execute the Report and Click the **Save** to save the Report.

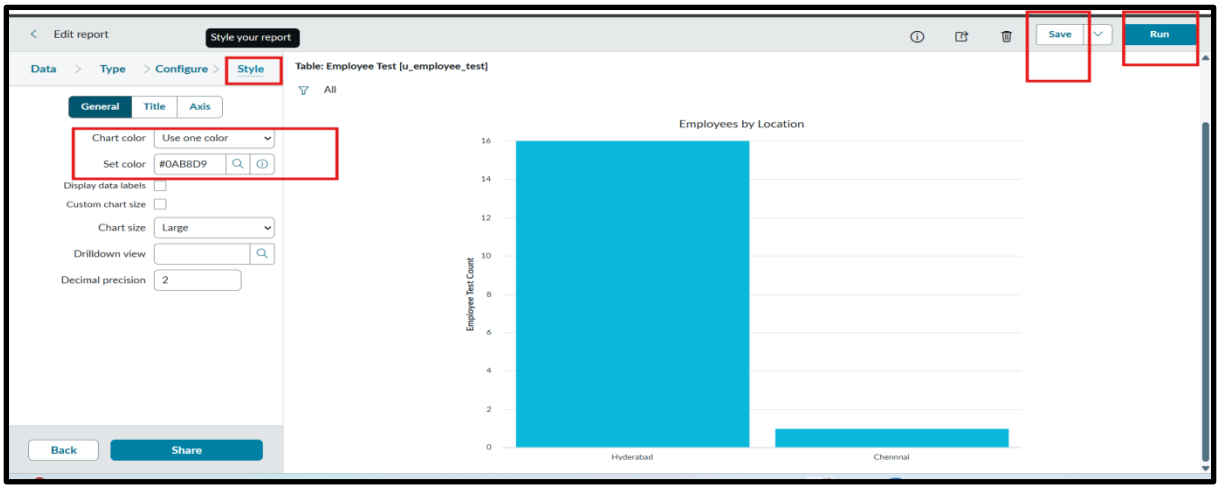

servicenow
All
Favorites
History
Workspaces
Admin
ServiceNow
Application scope: Global
Update set: E-Commerce Bot [Global]
Reports
FAVORITES
No Results
ALL RESULTS
Night Shift Summary Report
Night Shift Summary Reports
Platform Analytics Administrat...
Data Sources
Report Sources
Usage and governance
Reports
Report Statistics
Schedules
Scheduled Reports
Color Settings
Report Ranges
Properties
Reporting Properties
ServiceNow Studio
Build smarter and deploy faster
Access all the tools you need in one unified, intuitive development environment.
Open ServiceNow Studio
GO FURTHER
Power your workflow applications


Reports
Updated
Search
Actions on selected rows...
New
All
Title
Table
Type
Field Name
Created by
Updated
SLA Performance Trend
Incident [incident]
Bar
opened_at
admin
2026-01-02 02:32:06
Pending Justifications
Incident [incident]
List
admin
2026-01-02 01:13:00
Top Assignees with SLA Violations
Incident [incident]
Bar
assigned_to
admin
2026-01-01 22:54:58
Acknowledged vs Unacknowledged Breaches
Incident [incident]
Pie
u_sla_acknowledged
admin
2026-01-01 22:47:18
SLA Breaches by Assignment Group
Incident [incident]
Bar
assignment_group
admin
2026-01-01 22:23:49
SLA Breaches by Reason
Incident [incident]
Pie
u_sla_breach_reason
admin
2026-01-01 22:15:34
Compliant CI Trend
CMDB Health Scorecard [cmdb_health_scorecard]
Line
admin
2025-12-17 05:52:39


Edit report
Save
Run
Data
Type
Configure
Style
Report name
Employees by Location
Source type
Table
Table
Employee Test [u_employee_test]
Description
There is no description for this table. To add a description, please contact your admin.
Next
Table: Employee Test [u_employee_test]
All
Employees by Location
Employee Test Count
Hyderabad
Chennnai


Edit report
Style your report
Save
Run
Data
Type
Configure
Style
Filter the visualizations
Select one of the valid types
Bars
Compare individual or aggregate scores across data dimensions.
Pies and Donuts
Pies and Donuts show the proportions that make up a whole.
Back
Next
Table: Employee Test [u_employee_test]
All
Employees by Location
Employee Test Count
Hyderabad
Chennnai

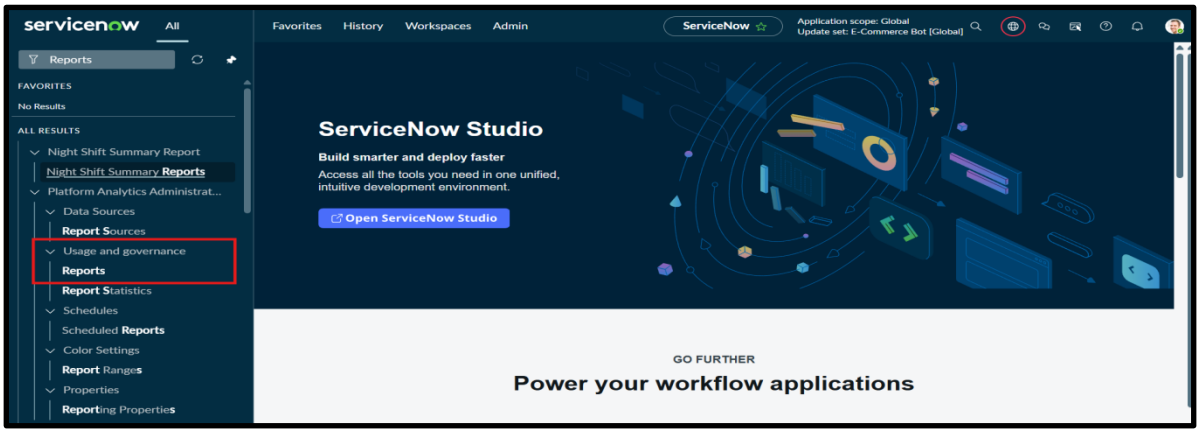

Edit report
Configure your report
Save
Run
Data
Type
Configure
Style
Group by
Location
Additional group by
Stack by
-- None --
Display data table
Configure function field
Aggregation
Count
Set Value Formatting
Max number of groups
System Default
Show Other
Back
Next
Table: Employee Test [u_employee_test]
All
Employees by Location
Employee Test Count
Hyderabad
Chennnai

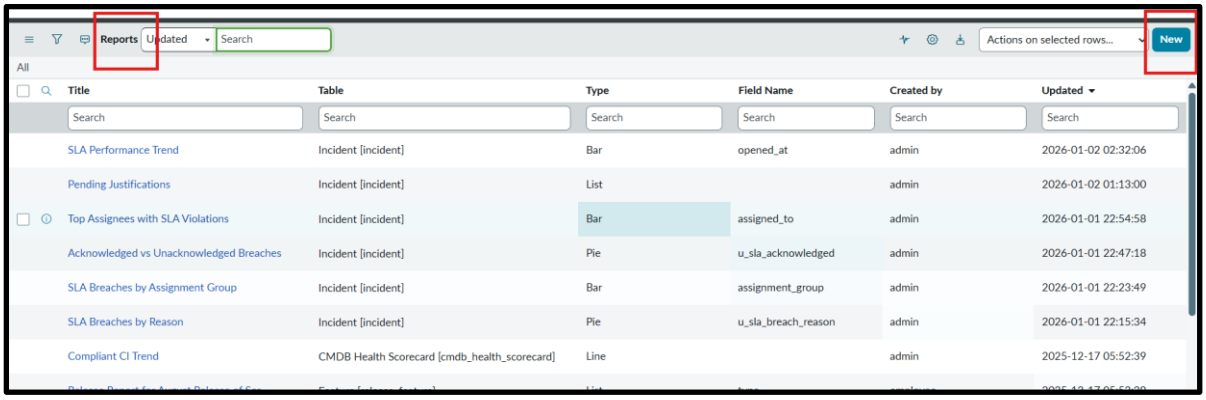


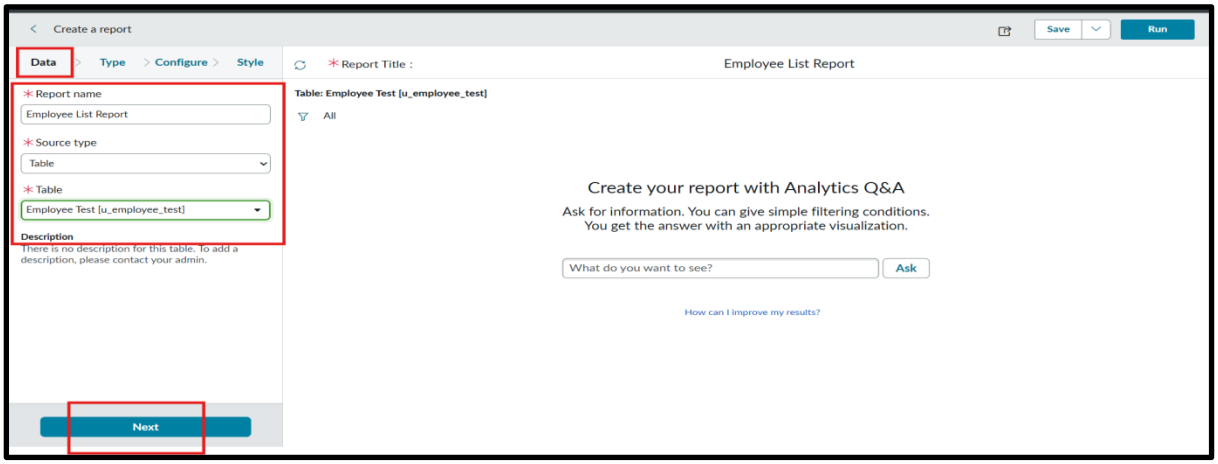
Report 3: Employee List Report.

Configuration

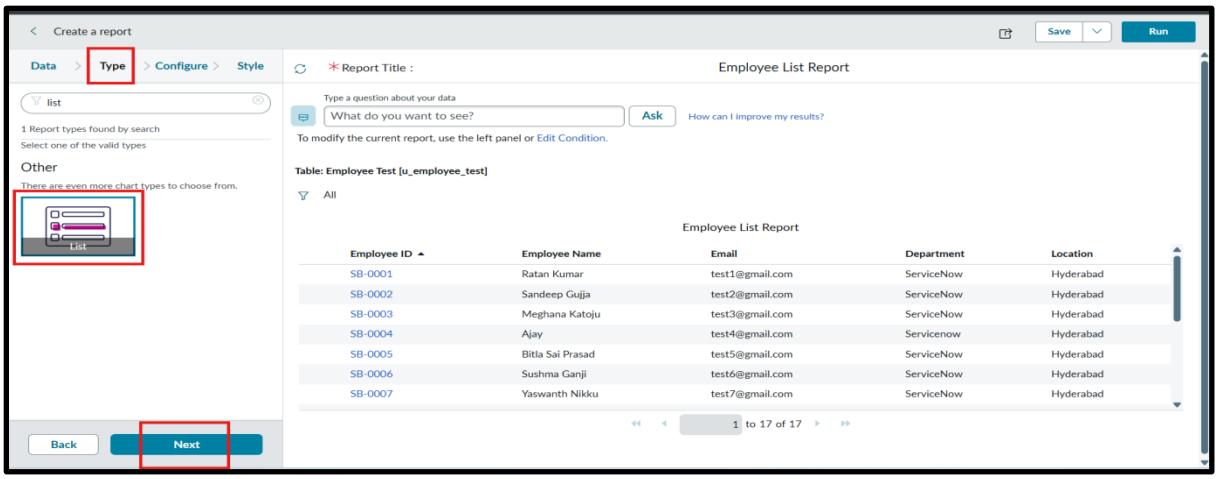
- Create a New Report name : **Employee List Report.**
- Table: **Employee Test.**
- Select Type: **List.**
- In Configure Select Columns:
    - **Employee ID.**
    - **Employee Name.**
    - **Email.**
    - **Department.**
    - **Location.**
- Now Select **next** and no required to choose anything in **Style**.

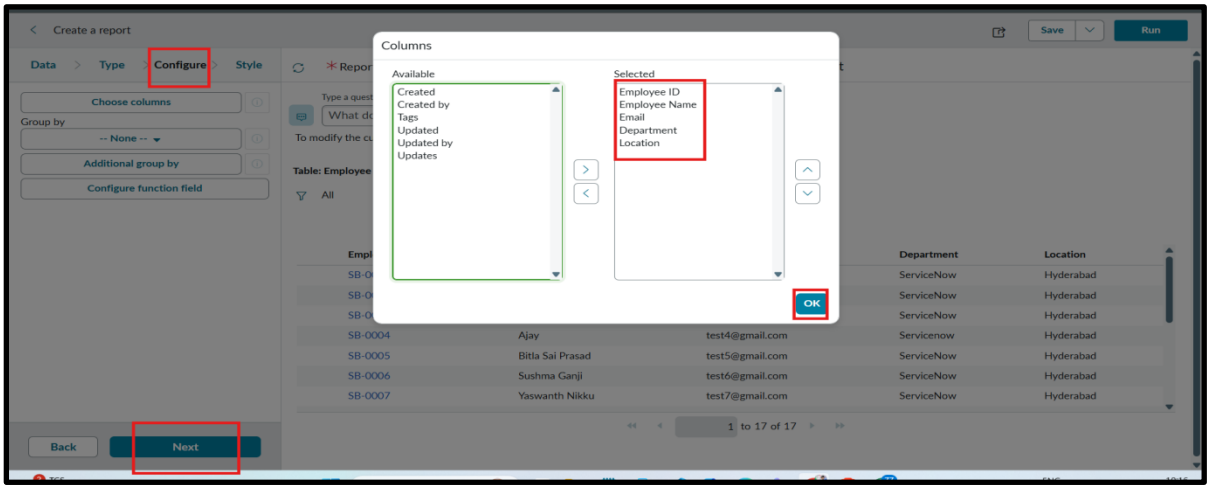


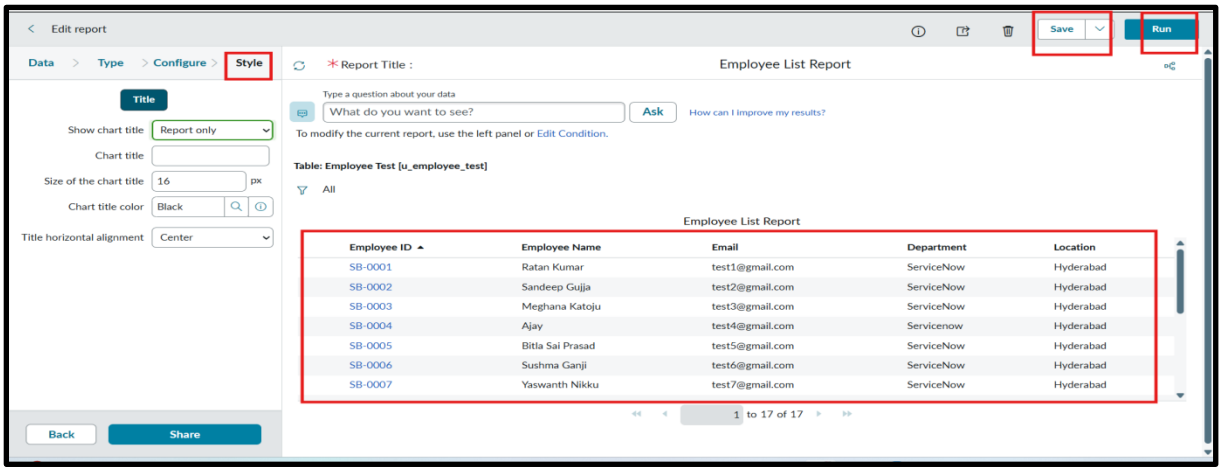




- Now Click **Run** to Execute the Report and Click **Save** to Save the Report.


Create a report
Save
Run
Data
Type
Configure
Style
Report Title :
Employee List Report
list
1 Report types found by search
Select one of the valid types
Other
There are even more chart types to choose from.
List
Type a question about your data
What do you want to see?
Ask
How can I improve my results?
To modify the current report, use the left panel or Edit Condition.
Table: Employee Test [u_employee_test]
All
Employee List Report
Employee ID
Employee Name
Email
Department
Location
SB-0001 Ratan Kumar test1@gmail.com ServiceNow Hyderabad
SB-0002 Sandeep Gujja test2@gmail.com ServiceNow Hyderabad
SB-0003 Meghana Katoju test3@gmail.com ServiceNow Hyderabad
SB-0004 Ajay test4@gmail.com Servicenow Hyderabad
SB-0005 Bitla Sai Prasad test5@gmail.com ServiceNow Hyderabad
SB-0006 Sushma Ganji test6@gmail.com ServiceNow Hyderabad
SB-0007 Yaswanth Nikku test7@gmail.com ServiceNow Hyderabad
1 to 17 of 17
Back
Next


Create a report
Save
Run
Data
Type
Configure
Style
Choose columns
Group by
-- None --
Additional group by
Configure function field
Columns
Available
Created
Created by
Tags
Updated
Updated by
Updates
Selected
Employee ID
Employee Name
Email
Department
Location
OK
Table: Employee
All
Department
Location
ServiceNow Hyderabad
SB-0004 Ajay test4@gmail.com Servicenow Hyderabad
SB-0005 Bitla Sai Prasad test5@gmail.com ServiceNow Hyderabad
SB-0006 Sushma Ganji test6@gmail.com ServiceNow Hyderabad
SB-0007 Yaswanth Nikku test7@gmail.com ServiceNow Hyderabad
1 to 17 of 17
Back
Next

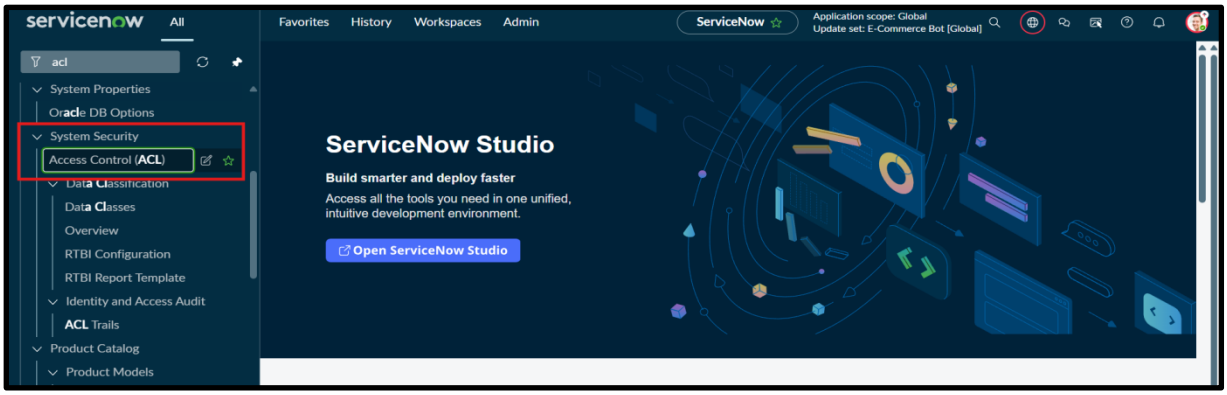

Edit report
Save
Run
Data
Type
Configure
Style
Report Title :
Employee List Report
Title
Show chart title
Report only
Chart title
Size of the chart title
16
px
Chart title color
Black
Title horizontal alignment
Center
Type a question about your data
What do you want to see?
Ask
How can I improve my results?
To modify the current report, use the left panel or Edit Condition.
Table: Employee Test [u_employee_test]
All
Employee List Report
Employee ID
Employee Name
Email
Department
Location
SB-0001 Ratan Kumar test1@gmail.com ServiceNow Hyderabad
SB-0002 Sandeep Gujja test2@gmail.com ServiceNow Hyderabad
SB-0003 Meghana Katoju test3@gmail.com ServiceNow Hyderabad
SB-0004 Ajay test4@gmail.com Servicenow Hyderabad
SB-0005 Bitla Sai Prasad test5@gmail.com ServiceNow Hyderabad
SB-0006 Sushma Ganji test6@gmail.com ServiceNow Hyderabad
SB-0007 Yaswanth Nikku test7@gmail.com ServiceNow Hyderabad
1 to 17 of 17
Back
Share

PHASE 9 : Add Reports to Dashboard

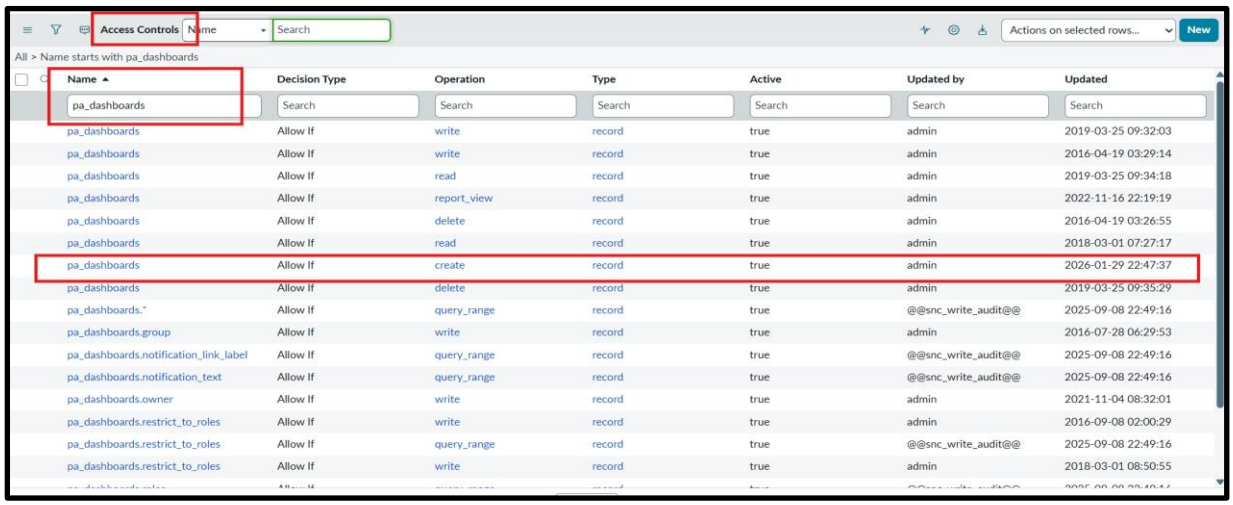
Create Dashboard

Navigation :

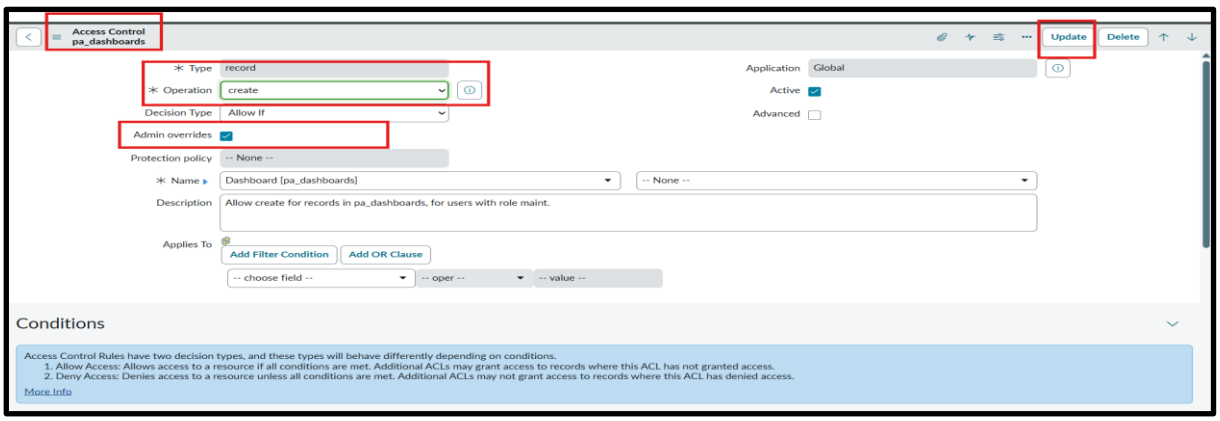
- Open All menus → Search for Dashboards.
- Under Self Service → Click Dashboards.
- On the top right corner Click Create Dashboards or /
- You can create Directly by using the backend name [PA_DASHBOARDS.FORM].
- Then Click New to Create new Dashboards.
- Only the backend name will open the form if the **ACL** has the **pa_dashboards** access to **Admin Overrides** True only.
- Then only we have access to the form.



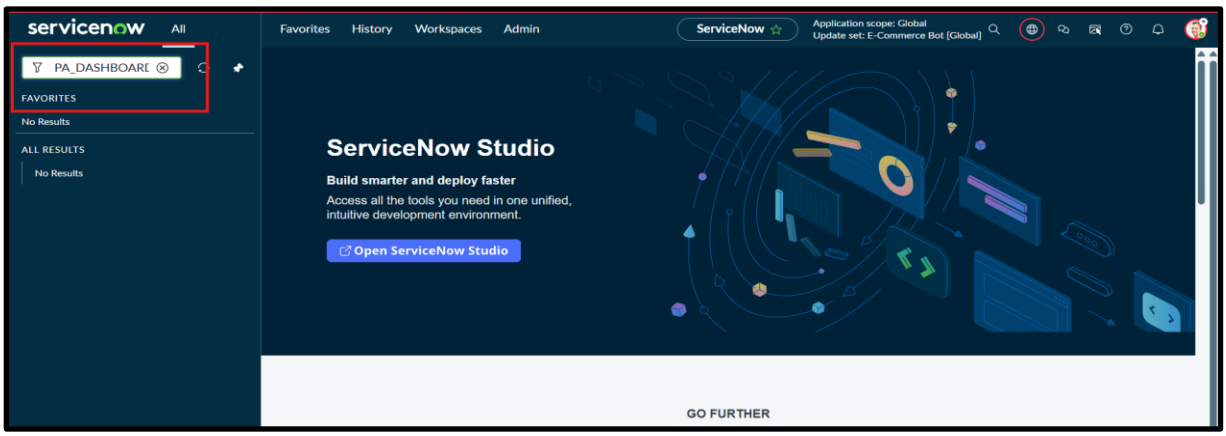
- After Opening the ACL Search the pa_dashboards on Name Column.

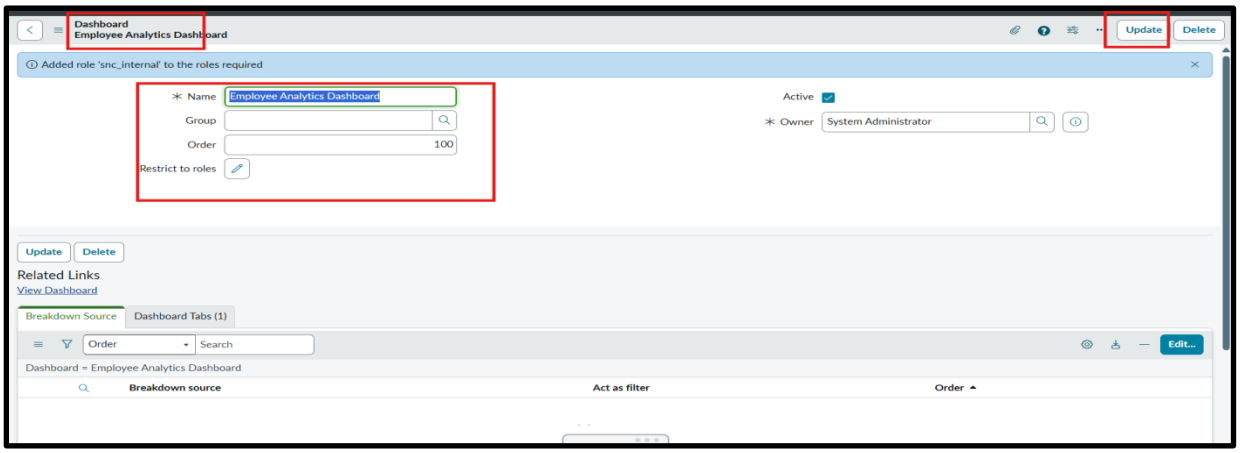


- Now Open the **Create Operation** row.



- Then Click Enable the True for Admin Overrides and Update the Form.
- Then the creation of dashboards will be possible by using the backend name.
- Now Open the **PA_DASHBOARDS.FORM** create dashboard.
- Dashboard name : **Employee Analytics Dashboards.**
- **Save** the Form.

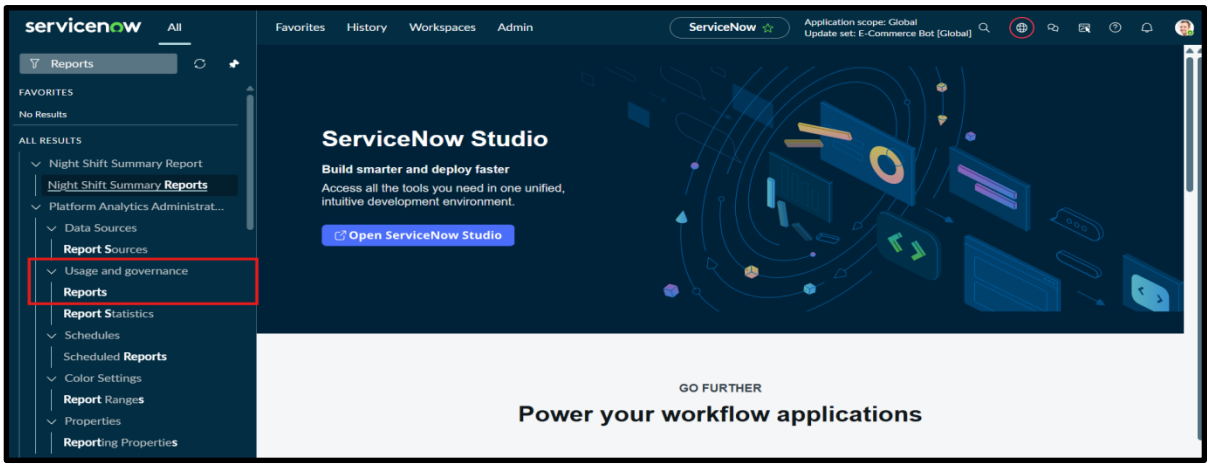


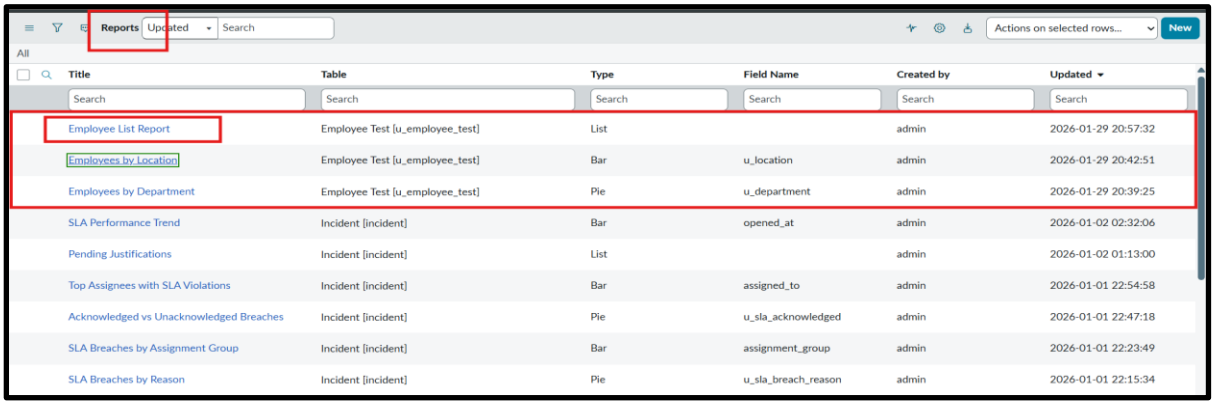


Add Reports to Dashboards.

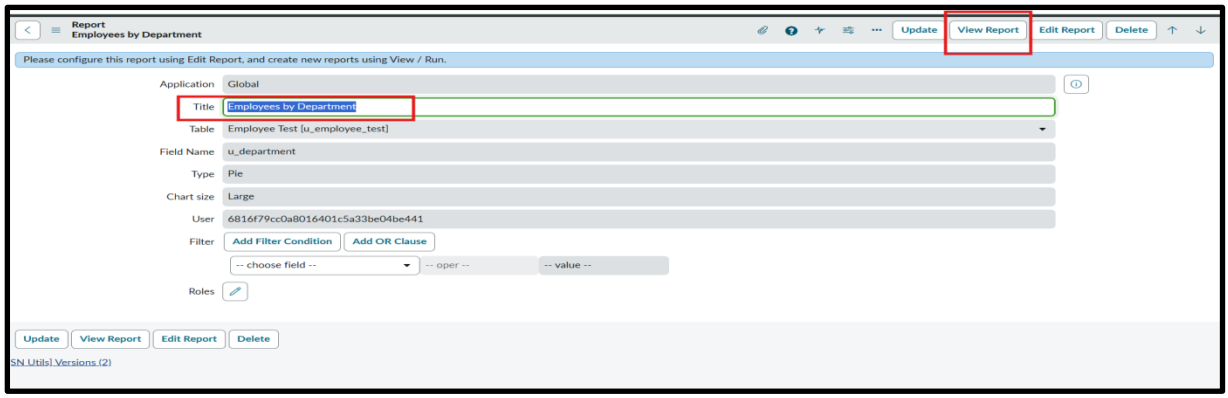
Navigation Steps:

- Open the **Previously Created Reports.**
- On All menus → type Reports Under **Usage and Governance** select **reports**.
- Open the **Employees by Department**.
- Select the **Edit Report** and Click the **Share** button then the right top corner Select the **Add to Dashboards** Option.
- Then Choose the Created **dashboard name** and Click **Add** button.
- Same Like that do the remaining **2 reports** in the same navigation how we added **Employees by Department.**
- The **Employees by Location** and **Employee List Report** these two reports are also added to the created dashboard.



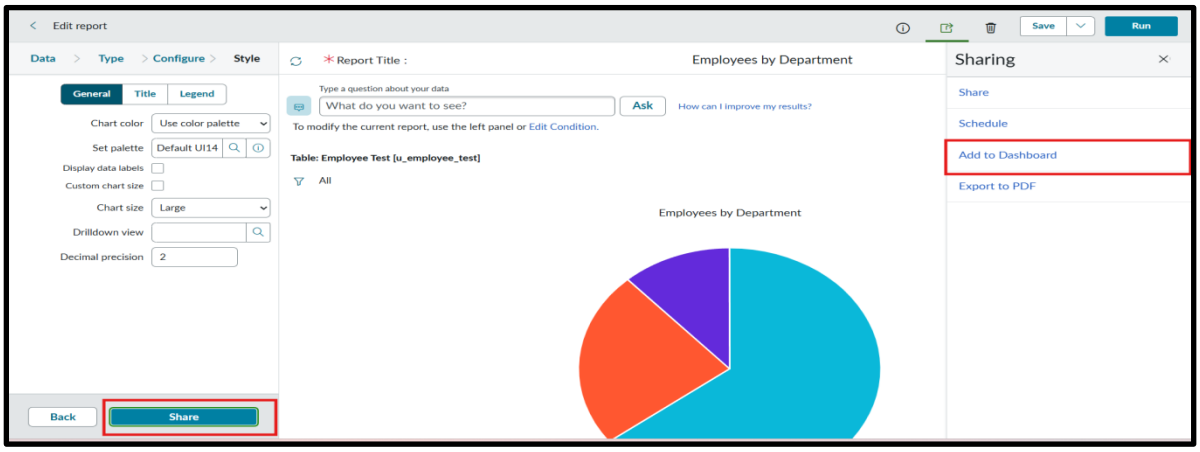


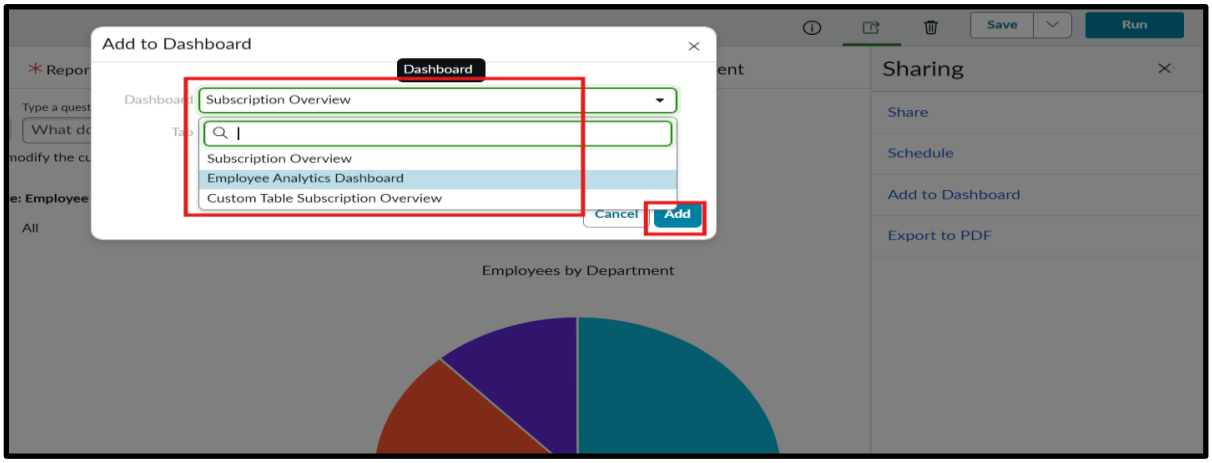
- After Opening the Report Click the **View Report** Button on the right side of



the Context menu.

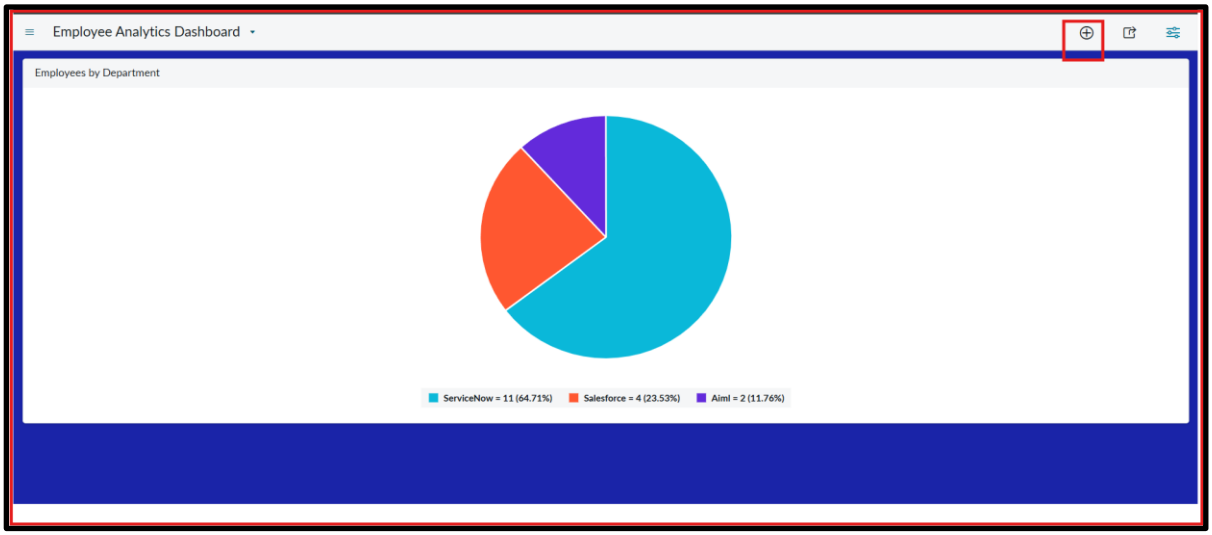
- After Clicking the View Report button, Now Click the **Share** button and then share options are visible on the right side.
- Now click the **Add to Dashboards**.



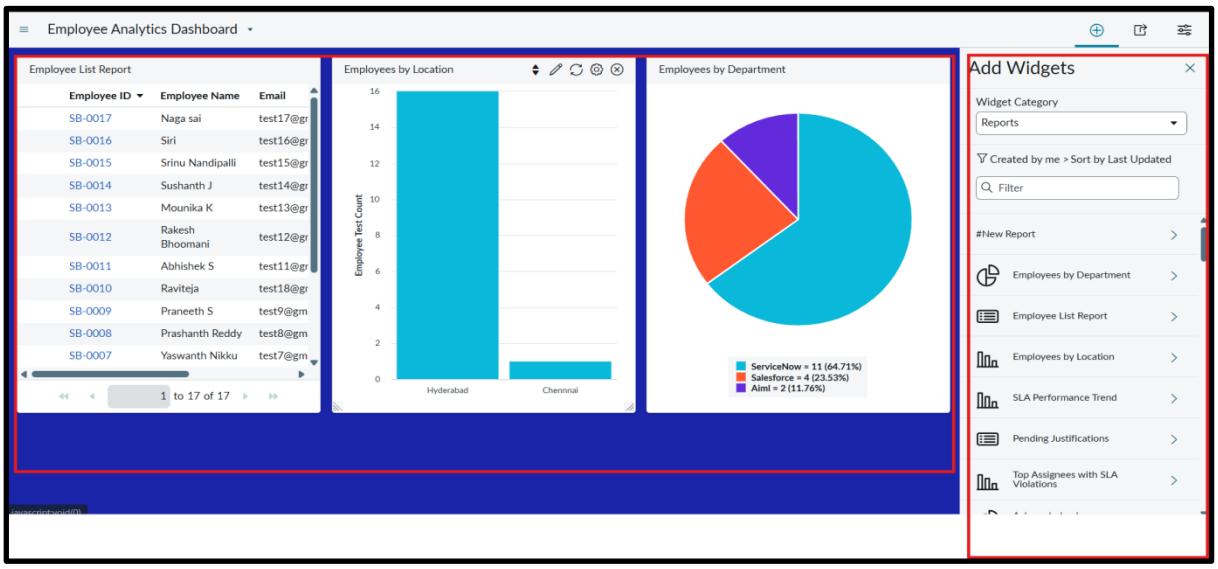


- Now Add the Dashboard that you created called **Employee Analytics Dashboards.**

- Then Click **Add** button.
- Now the **Report** is added to the **dashboards**.
- Now, Same in the process, add the remaining **two reports** to the **dashboard**.



- From the + icon also we add widgets that our reports from there.
- If we want we can also configure the background color.





- Here the three Reports are added successfully by varying **Department , Location and List Report.**
- We can also restrict the **Users,Groups and Roles** From Share Option.

Conclusion:

1. This project successfully implemented an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency, and data integrity throughout the employee onboarding and update process.
2. A key enhancement in this project was the use of Coalesce in the Transform Map. By enabling Coalesce on the Employee ID field, the system intelligently identifies existing employee records during every import. This prevents the creation of duplicate records and ensures that repeated imports update existing data instead of inserting redundant entries. As a result, the employee database remains clean, reliable, and consistent, which is critical in real-world enterprise environments where data may be imported multiple times from external systems.
3. In addition to data integrity, the project was extended with Reports and Dashboards to provide meaningful insights and real-time visibility into employee data and import activities. Custom reports were created to track:

    - Total employee records.
    - Newly added employees.
    - Updated employee records.
    - Department-wise employee distribution.
    - Import success and error counts.
4. These reports were then consolidated into a Dashboard, allowing administrators and HR teams to monitor employee data trends, import performance, and data quality from a single centralized view. Dashboards improved decision-making by presenting information visually through charts and graphs, reducing the need for manual analysis.